

# Finite-contour cocycles for (quasi-)periods and $L$ -functions of meromorphic modular forms on $\mathrm{SL}_2(\mathbb{Z})$

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## Abstract

In this note, we synthesize and generalize previous work, which (1) extracted periods for weak cusp forms in a manifestly finite manner [1], (2) identified a non-holomorphic regularization prescription to define finite analogs of L-functions associated to weakly holomorphic modular forms [2], and (3) identified a contour prescription to define an L-function for meromorphic modular forms with poles inside the fundamental domain [3]. Our construction, derived from the cohomological period extraction of weak cusp forms in [1], shows that these three lines of work not only naturally align, but derive from the *same* manifestly finite linear functional acting on the space of  $\mathrm{SL}_2(\mathbb{Z})$  modular forms.

We call this linear functional the  $L$ -integral, and denote it  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ ; here  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$  are paths from  $S^{-1}\tau_0, T^{-1}\tau_0$ , respectively, to  $\tau_0$ . When  $f \in M_k$ ,  $L^*(f, s) = \Gamma(s) \sum_{n \neq 0} \frac{c_f(n)}{(2\pi n)^s}$ : the classical L-function. Our L-integrals are finite,  $\tau_0$ -independent, and path independent within each homotopy class (if  $f$  has poles at  $0 < \mathrm{Im}(\tau) < \infty$ ), for any meromorphic modular form  $f$ .

We use this L-integral to (1) extract published numerical values of the periods of the weak cusp form  $\widehat{\Delta} := \frac{1}{q} + O(q^2) \in S_{12}^!$  in [1], (2) recover the regularized expressions for the divergent Mellin transform of  $f$  in [2], and (3) recover the L-functions of [3] for meromorphic forms with interior poles that are regular at the cusps. When  $f$  has poles inside the fundamental domain,  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is unique and constant for paths within the same homotopy class; related  $L^*(f, s)$ -integrals along paths in different homotopy classes differ by wall-crossing residues. Finally, all major results were tested to arbitrary (50 digit) precision in our python scripts [4].

## 1 Introduction and main results

In this paper, we introduce a general framework that defines L-functions for generic modular forms on the full group  $\mathrm{SL}_2(\mathbb{Z})$ , and extracts their periods for the explicit case when the holomorphic cusp forms are one-dimensional, i.e. when  $\dim(S_k) = 1$ . Classically, L-functions and periods are both well-defined and well-studied for holomorphic modular forms. However, there is a growing body of work exploring periods [1] and L-functions [2] for weakly holomorphic modular forms. Similarly, there are natural physical motivations to consider L-functions for properly meromorphic modular forms, which have poles *within* the fundamental domain [3].

In these works, and specifically in [2, 3], and throughout physics, divergences due to singularities in quantities of interest necessitate the introduction of regulators in order to define finite quantities. Regulators, though, often break symmetries, and care is needed to check that the regulator has not injected undue violence to the system. Of relevance here is that [2] introduced a non-holomorphic regulator to deal with the singularities at either end of the classical Mellin contour,  $\int_0^\infty$ , sufficient to regulate divergences at the cusps, but which violates holomorphy in the deformed contour-integral, ruining path-independence. Simultaneously, [3] concerned itself with L-functions for modular forms which diverge within the interior of the fundamental domain. Here, holomorphic dependence on  $\tau$  is paramount, which puts this treatment at odds with the regulator in [2].

In this note, we introduce manifestly finite contour integral functionals to define L-functions for generic meromorphic modular forms that are of sufficient generality to handle any combination of poles at cusps and poles within the fundamental domain. Our construction is motivated by, and makes precise contact with, work which extracts period-coefficients from weak cusp forms [1]. Our integrals are manifestly finite throughout, and reproduce all results in the literature, unifying the quasi-period results and the L-function extensions within one concrete, natural framework.

Throughout,  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ , and we consider modular forms on  $\Gamma$  of even integer weight and trivial multiplier system. Write  $M_k$  for the holomorphic modular forms of weight  $k$ ,  $S_k \subset M_k$  for the cusp forms, and  $M_k^!$  for the weakly holomorphic forms (holomorphic on  $\mathbb{H}$ , with at most a pole at the cusp). Write  $F_k$  for the *meromorphic modular forms* of weight  $k$ : the weight- $k$  part of the field of fractions of the graded ring  $M_* = \bigoplus_k M_k$ , i.e. the quotients  $f = f_1/f_2$  with  $f_i \in M_{k_i}$ ,  $f_2 \neq 0$ , and  $k = k_1 - k_2$ ; equivalently, the weight- $k$  functions meromorphic on  $\mathbb{H}$  and at the cusp. Thus  $M_k \subset M_k^! \subset F_k$ . Finally,  $S_k^!$   $\supset S_k$  denotes the *weakly holomorphic cusp forms*: those  $f = \sum_{n \in \mathbb{Z}} c(n) q^n \in M_k^!$  with vanishing constant term,  $c(0) = 0$ . We write  $E_k = 1 + O(q) \in M_k$  for the unit-normalized Eisenstein series,  $\Delta := (E_4^3 - E_6^2)/1728 = q + O(q^2) \in S_{12}$  for the modular discriminant, and  $j = 1728 E_4^3/(E_4^3 - E_6^2) = 1/q + O(q^0) \in M_0^!$  for the Hauptmodul of  $\mathrm{SL}_2(\mathbb{Z})$ .

In a nutshell, this note is the natural extension of bridging the gap between the manifestly finite extraction of quasi-periods of  $\widehat{\Delta}$  in [1] and the manifestly regulator-dependent derivation of effective L-functions for weak cusp forms in [2]. By leveraging the construction underlying the computation in [1], we uncover a unified, manifestly finite formalism to define L-functions, and extract L-values which define period polynomials, for truly generic meromorphic modular forms.

## 1.1 Manifestly finite extractions of period polynomials for weak cusp forms

To understand the manifestly finite extraction of the quasi-periods in [1], we begin by studying the manifestly  $\mathrm{SL}_2(\mathbb{Z})$ -covariant differential,

$$d\omega := d\tau f(\tau)(X - \tau Y)^{k-2}, \quad (1.1)$$

where  $f$  is a modular form of weight  $k$  on  $\mathrm{SL}_2(\mathbb{Z})$ , and contour integrals of  $P(X, Y) := \int_a^b d\omega$  yield polynomials  $P \in V_{k-2} := \mathbb{C}[X, Y]_n$  which have natural action under  $\mathrm{SL}_2(\mathbb{Z})$ , i.e. for each  $\gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$ ,  $(f|_\gamma)(\tau) = (c\tau + d)^k f(\tau)$  and  $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$ .

Period polynomials of holomorphic cusp forms  $f \in S_k$ , which are commonly written as

$$r_f(X, Y) = \int_0^{i\infty} f(\tau)(X - \tau Y)^{k-2} d\tau, \quad (1.2)$$

are clearly integrals of the  $\mathrm{SL}_2(\mathbb{Z})$ -covariant differential over a contour stretching between the cusp at  $\tau = 0$  and its S-image. Yet, one may choose a different contour, at the expense of introducing more general machinery.

The machinery needed to define period polynomials as contour integrals over more general contours is essentially ignorable in the case of holomorphic cusp forms. However, it is essential to being able to cleanly extend the notion of a modular form's period polynomial and L-functions beyond of the narrow, but important, context of holomorphic cusp-forms. For instance, relaxing from a holomorphic cusp form to a weakly holomorphic cusp form introduces essential singularities at *both* ends of the classical integration contour.

Thus, we seek to understand period polynomials even of holomorphic cusp forms via contour integrals between more generic end-points. Fix the even weight  $k$ , set  $n := k - 2$ , and let  $V_n =$

$\mathbb{C}[X, Y]_n$  carry the weight- $n$  action  $(P|_\gamma)(X, Y) = P(aX + bY, cX + dY)$  of  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$ . The weight- $k$  modularity of  $f$  makes the  $V_n$ -valued one-form

$$\omega_f := f(\tau)(X - \tau Y)^n d\tau \quad (1.3)$$

invariant under the simultaneous action of  $\Gamma$  on  $\tau$  and on  $(X, Y)$ .

The object we are really after is a *primitive* of  $\omega_f$ . Since  $\omega_f$  is closed on  $\mathbb{H}$  away from the poles of  $f$ , every simply-connected pole-free region carries an antiderivative  $F$  with  $dF = \omega_f$  — a  $V_n$ -valued function, determined up to an additive  $V_n$ -constant. Were  $F$  itself  $\Gamma$ -invariant,  $f$  would have no periods; it is not, and that failure is the entire subject. (The primitive  $F$  is the *indefinite* integral of  $\omega_f$ , a transcendental function of  $\tau$ ; the period polynomial we ultimately extract is a *definite* one — a difference of  $F$  across a contour, hence a fixed element of  $V_n$ , the polynomial *defect* of  $F$ 's modularity rather than a primitive itself.) We record the failure one group element at a time and — the point of the whole construction — *between finite interior points of  $\mathbb{H}$ , never at a cusp*:

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1}(F(\tau_0) - F(\gamma^{-1}\tau_0)) = (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f, \quad \gamma \in \Gamma. \quad (1.4)$$

The right-hand equality makes  $C_\gamma^f(\tau_0)$  manifestly finite and independent of which primitive  $F$  was chosen (the additive constant cancels in the difference); the middle one makes its meaning plain —  $C_\gamma^f(\tau_0)$  is the primitive's failure to be  $\Gamma$ -modular, read at the basepoint  $\tau_0$ . That basepoint is where we *anchor* the primitive: it is the finite, convergent stand-in for the cusp at which the classical Eichler integral is silently anchored, recovered in the holomorphic case as  $\tau_0 \rightarrow i\infty$  (§2). The map  $\gamma \mapsto C_\gamma^f(\tau_0)$  satisfies the cocycle relation  $C_{gh}^f = C_g^f|_h + C_h^f$  (Lemma 2.1).

Since  $\Gamma = \langle S, T \rangle$ , the cocycle is determined by its two generator values, and it is natural to single-out (homotopy classes of) S- and T-contours associated with the base-point  $\tau_0$ :

**Definition 1.1** (S- and T-segments at a basepoint). *Let  $\tau_0 \in \mathbb{H}$ , and consider any  $\sigma \in \mathrm{SL}_2(\mathbb{Z})$ . We denote the path from  $\sigma^{-1}(\tau_0) \rightarrow \tau_0$  by  $\gamma_{\tau_0}^\sigma$ . Further, let  $f$  be a meromorphic modular form of weight  $k \in 2\mathbb{Z}$  on  $\mathrm{SL}_2(\mathbb{Z})$ , which may have poles at cusps and may have poles within the interior of the upper half-plane. Now, consider  $\sigma \in \{S, T\} = \left\{ \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \right\}$ . Here,  $\gamma_{\tau_0}^\sigma$  is a smooth contour,*

$$\begin{cases} \text{S-segment:} & \gamma_{\tau_0}^S : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^S(0) = -1/\tau_0, \gamma_{\tau_0}^S(1) = \tau_0, \\ \text{T-segment:} & \gamma_{\tau_0}^T : [0, 1] \rightarrow \mathbb{H}, & \gamma_{\tau_0}^T(0) = \tau_0 - 1, \gamma_{\tau_0}^T(1) = \tau_0, \end{cases} \quad (1.5)$$

where both contours avoid the  $\Gamma$ -orbit of the poles of  $f$  in  $\mathbb{H}$ .

All contours in this note lie between finite interior points of  $\mathbb{H}$ , and all contour avoids singular points. Finiteness of every integral is therefore structural; there is never anything to regulate.

The price of trading one divergent contour for a family of finite ones is *redundancy*: each  $C_\gamma^f(\tau_0)$  carries both a group element  $\gamma$  and an auxiliary basepoint  $\tau_0$ , whereas the period polynomial must refer to neither. The two carry-alongs are not on the same footing. The  $\gamma$ -dependence is the data — it is the cocycle. The  $\tau_0$ -dependence is pure redundancy, and the primitive picture names it exactly:  $\tau_0$  is where we anchored  $F$ , so moving it re-anchors the primitive and, by (2.10), shifts the *whole* family  $\{C_\gamma^f\}_\gamma$  by one and the same polynomial  $P = (2\pi i)^{n+1}(F(\tau_0) - F(\tau_0'))$  — the same  $P$  for every  $\gamma$ , which is what lets a quotient remember anything at all.

Degree-one group cohomology is the standard bookkeeping for exactly this, and three definitions suffice. A 1-*cocycle* is a function  $c: \Gamma \rightarrow V_n$  with  $c_{gh} = c_g|_h + c_h$ , and  $\gamma \mapsto C_\gamma^f(\tau_0)$  is one for each

fixed  $\tau_0$  (Lemma 2.1). A 1-coboundary is the special cocycle  $\delta P: \gamma \mapsto P|_\gamma - P$  from a polynomial  $P \in V_n$  — precisely the re-anchoring shift above. The cohomology class  $[C^f] \in H^1(\Gamma; V_n) := \{\text{cocycles}\}/\{\text{coboundaries}\}$  is what survives forgetting the anchor: it no longer remembers  $\tau_0$ , and a representative of it is any single cocycle  $C^f - \delta P$ . This is half the journey from  $C_\gamma^f(\tau_0)$  to  $r_f$  — forget the basepoint by passing to the class. The other half — pinning one representative, which is what turns a class back into a polynomial — is forced by the remaining structure of  $\Gamma$ , and which generator does the pinning is the subject of the next two paragraphs.

A reader outside the field might object that no machinery should be needed: the classical period polynomial is a single integral along one  $S$ -contour, so why not take the finite  $S$ -contour  $C_S^f(\tau_0)$  and stop? Because at a finite basepoint the raw  $S$ -value is pure redundancy. Torsion alone forces  $C_S^f|(1+S) = C_{S^2}^f = C_{-I}^f = 0$  at every  $\tau_0$ , and since  $S$  acts on  $V_n$  as an involution, its  $(-1)$ -eigenspace is also the image of  $\delta_S$ : the value  $C_S^f$  lies in  $\ker(1+S) = \text{Im } \delta_S$ , exactly the space of shifts that basepoint motion induces on it. (At  $k=12$  this kernel is six-dimensional inside the eleven-dimensional  $V_{10}$ ; membership selects a subspace, never a point.) The two relations cutting out the Kohnen–Zagier space  $W_{k-2}$  [12] therefore have different statuses: the  $S$ -relation is *free*, holding before any choice is made, while the  $U$ -relation must be *purchased* — it appears only after the representative is pinned, and pinning is the role of the other generator.

The generator  $T$  is built for the job. It has infinite order and acts on  $V_n$  unipotently — a single Jordan block — so  $\delta_T := (\cdot)|_T - \text{id}$  has one-dimensional kernel and cokernel: the equation has one scalar of obstruction and one scalar of slack. We choose, within the class, the representative  $C^f - \delta P$  whose  $T$ -value vanishes, i.e. solve  $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$  on  $V_n$ . Solvability is itself arithmetic: unit-segment  $T$ -periodicity identifies the obstruction in  $V_n/\text{Im } \delta_T \cong \mathbb{C} \cdot [X^n]$  as the constant Fourier mode (Lemma 3.1),  $[C_T^f]_{X^n} = (2\pi i)^{n+1} c_f(0)$  — the residue of  $\omega_f$  at the very cusp that  $T$  stabilizes. The normalized representative exists if and only if  $c_f(0) = 0$ ; classes that pass are the *parabolic* ones, identified by Eichler–Shimura [12] with  $S_k \oplus \bar{S}_k$ , the lone obstructed direction being Eisenstein. Cuspidality, in this framework, is not a convergence hypothesis; it is the solvability condition for the normalization.

With the  $T$ -value normalized away, the entire cuspidal content sits on the surviving generator value,

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P), \quad (1.6)$$

well-defined up to the line  $\mathbb{C}(X^n - Y^n)$  left over from  $\ker \delta_T = \mathbb{C}Y^n$  — an ambiguity touching only the boundary monomials, which is why Theorem 1.4 ranges over the interior  $\ell$  alone. The solved  $P$  is itself basepoint-dependent — re-anchoring at  $\tau'_0$  sends it to  $P + (F(\tau_0) - F(\tau'_0))$  modulo  $\mathbb{C}Y^n$  — but that increment is the *same*  $F(\tau_0) - F(\tau'_0)$  that shifts  $C_S^f$  under (2.10), so the two move in lockstep and cancel in (1.6): the interior coefficients of  $r_f$  are  $\tau_0$ -independent (Corollary 3.8), and only the boundary line  $\mathbb{C}(X^n - Y^n)$  can shift with the anchor. The  $U$ -relation now arrives, purchased as promised: conditioned on the vanishing  $T$ -value, the group relation  $(ST)^3 = -I$  becomes  $r_f|(1+(TS)+(TS)^2) = 0$ , and  $r_f$  is the period polynomial. Indeed, for holomorphic  $f \in S_k$ , sending  $\tau_0 \rightarrow i\infty$  kills  $C_T^f$  outright and (1.6) collapses to the classical integral (1.2) — whose familiar freedom from  $\tau_0$  and  $\gamma$  is thereby revealed as the silent choices  $\gamma = S$ ,  $\tau_0 = i\infty$ . Its interior coefficients are the periods against Brown’s rational basis,  $[r_f]_{X^{k-2-\ell}Y^\ell} = \omega^\pm(f) c_S^\ell$ ; the precise coordinate statement is given in Section 2. The finite  $S$ -contour was the right object all along, but it requires the correction term in (1.6), sourced entirely by the  $T$ -contour — and producing that correction in closed form is the business of this paper.

The coboundary equation admits a closed-form inverse — the polynomial inversion of [7] — under which the correction  $-(P|_S - P)$  re-expresses itself as a second contour integral, along the  $T$ -cycle, against an explicit  $\mathbb{Q}$ -rational kernel, one for each interior monomial of  $r_f$ . As we show in

Section 3, these kernels are the values, at the interior integer points of the critical strip, of a single difference of Hurwitz zeta functions — and the Hurwitz difference, unlike its integer shadows, is defined at every  $s \in \mathbb{C}$ . The following two definitions are very helpful:

**Definition 1.2** (Hurwitz T-kernel). *Let  $\zeta(s, x)$  denote the classical Hurwitz zeta-function,  $\zeta(s, x) := \sum_{n \geq 0} (n + x)^{-s}$ . We define the Hurwitz T-kernel as,*

$$\tilde{k}_T(\tau, s) := \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)} \zeta(s - (k - 1), \tau + 1). \quad (1.7)$$

**Definition 1.3** (Cuspidal basis coefficients,  $\dim S_k = 1$ ). *Let  $k \in 2\mathbb{Z}$  with  $\dim S_k = 1$ , and let  $f \in S_k^!$ . For  $\ell \in \{0, 1, \dots, k - 2\}$ , let*

$$c_S^\ell := [p_f^\pm]_{X^{k-2-\ell} Y^\ell} \in \mathbb{Q} \quad (1.8)$$

*denote the coefficient of  $X^{k-2-\ell} Y^\ell$  in the cuspidal basis element  $p_f^\pm$  of the period polynomial space  $W_{k-2}^\pm$  [12], in Brown's  $S$ -cocycle normalisation [1].*

With these definitions, we may compactly express the two period coefficients  $\omega^\pm(f)$ , which determine the period polynomial of  $f \in S_k^!$  when  $\dim(S_k) = 1$ , via the following theorem:

**Theorem 1.4** (Periods of  $f \in S_k^!$  as an explicit linear functional, when  $\dim S_k = 1$ ). *Let  $k, f \in S_k^!$ , and  $c_S^\ell$  be as in Definition 1.3, and let  $\ell \in \{1, \dots, k - 3\}$ , so that  $s = \ell + 1$  ranges over the interior integer points  $\{2, \dots, k - 2\}$  of the critical strip. Then the period coefficients  $\omega^\pm(f)$  are given by the two-segment linear functional,*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^\ell \binom{k-2}{\ell}}{c_S^\ell} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^\ell d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell + 1) d\tau \right]. \quad (1.9)$$

*This functional is independent of  $\tau_0 \in \mathbb{H}$  and of the admissible contours  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ . Its value depends on  $\ell$  only through its parity, which selects the extracted coefficient:  $\omega^+(f)$  for  $\ell$  even, and  $\omega^-(f)$  for  $\ell$  odd.*

The structure of (1.9) is exactly that of the normalized cocycle: the bare Mellin monomial on the  $S$ -cycle, plus the normalization term on the  $T$ -cycle. The second integral is not a second physical contribution; it is the coboundary subtraction written as a contour integral.

At  $k = 12$  the canonical test pair is  $f \in \{\Delta, \hat{\Delta}\}$ , where  $\hat{\Delta} := 1/q + \mathcal{O}(q^2) \in S_{12}^!$  is the unique weakly holomorphic cusp form with this expansion; explicitly,  $\hat{\Delta} = (j^2 - 1464j + 142236)\Delta$ . Applying (1.9) to  $\Delta$  recovers Brown's values of  $\omega^\pm(\Delta)$ ; applying it to  $\hat{\Delta}$  produces the quasi-periods  $\eta^\pm(\hat{\Delta})$  from a finite-endpoint contour pair, with no divergence at the cusp. It is straightforward to see that our linear functional applies to any  $k$  for e.g.  $\dim S_k \geq 2$ . Here, the corresponding statements become matrix identities. Working out the explicit matrix-valued kernels for  $\dim(S_k) \geq 2$ , and connecting them to the existing Eichler–Shimura matrix-period basis literature, is a natural generalization of Theorem 1.4. We do not pursue this generalization, here.

## 1.2 Manifestly finite L-functions for generic, meromorphic modular forms

Nothing in Definition 1.2, however, requires  $s$  to be an integer — and this is where the  $L$ -function arrives for free. Promoting the Mellin monomial  $\tau^{\ell-1}$  to  $\tau^{s-1}$  on the  $S$ -cycle and the Bernoulli pair to its Hurwitz parent on the  $T$ -cycle defines the functional at the heart of this note:

**Definition 1.5** (L-integral of a meromorphic modular form). *Let  $f \in F_k$  be a meromorphic modular form of weight  $k \in 2\mathbb{Z}$  on  $\mathrm{SL}_2(\mathbb{Z})$ , and let  $\tau_0 \in \mathbb{H}$ . Using the contour in Definition 1.1 and the Hurwitz  $T$ -kernel in Definition 1.2, we define the L-integral of  $f$  to be,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right]. \quad (1.10)$$

Both integrals are finite for every admissible contour pair and every  $s \in \mathbb{C}$ , by the same structural mechanism as before: compact contours in the interior of  $\mathbb{H}$ . No regulator, asymptotic-subtraction argument, or analytic-continuation step is required at any  $s$ . And the construction certifies itself: differentiating (1.10) in  $\tau_0$  leaves only endpoint terms, which the modularity of  $f$  collapses to  $f(\tau_0)$  multiplied by precisely the combination that the unit-shift equation defining  $\tilde{k}_T$  sets to zero. The identity that defines the kernel is the identity that makes the functional basepoint-independent: normalizability of the cocycle and well-definedness of the  $L$ -function are the same statement.

The specializations of (1.10) now recover the classical  $L$ -function and each of its partial extensions as evaluations of one finite functional. For holomorphic cusp forms, the cusp limit  $\tau_0 \rightarrow i\infty$  sends the  $S$ -segment to the classical Mellin contour while the  $T$ -segment vanishes exponentially: the L-integral is the Hecke–Weil completed  $L$ -function. (At finite  $\tau_0 = i\Lambda$ , the  $S$ -segment alone is the  $\Lambda$ -truncated Mellin contour of [3]; the  $T$ -segment is the missing piece that completes it to a cocycle and renders the  $\Lambda \rightarrow \infty$  limit unnecessary.) For weakly holomorphic  $f$ , where the cusp limit is genuinely pathological, the opposite specialization is available: at the elliptic fixed point  $\tau_0 = i$  of  $S$ , the  $S$ -segment collapses to a point, and a single unfolding of the surviving compact  $T$ -integral reproduces the incomplete-gamma sums of [2] term by term. Their non-holomorphic-regulator prescription is thereby revealed as one closed-form evaluation of the finite functional — not an independent definition — and, because the functional is  $\tau_0$ -independent and finite on all of  $M_k^!$ , this identifies the BFK candidate as the canonical  $L$ -function of any weak modular form.

**Theorem 1.6** ( $L$ -function for weakly holomorphic forms). *Let  $k \in 2\mathbb{Z}$  and  $f \in M_k^!$ . For every  $\tau_0 \in \mathbb{H}$  and every  $s \in \mathbb{C}$ , the  $L$ -integral  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  of (1.10) is finite and independent of both  $\tau_0 \in \mathbb{H}$  and of the admissible contour choices  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ . We write  $L^*(f, s)$  for the common value. Specializing to  $\tau_0 = i$ , the  $S$ -integral in (1.10) vanishes, and  $L^*$  reduces to a single integral on the compact contour  $[i-1, i] \subset \mathbb{H}$ , with closed-form sum*

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[ \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (1.11)$$

*The right side of (1.11) is the completed  $L$ -function  $L^*(f, s)$  of  $f$ . These  $L$ -integrals reproduce the Hecke–Weil Mellin transform for holomorphic modular forms, as well as reproduce extant formulae from the literature for weakly holomorphic modular forms [2].*

We unpack how Theorem 1.6 naturally reproduces the Hecke–Weil Mellin transform for holomorphic modular forms, as well as reproduce extant formulae from the literature for weakly holomorphic modular forms [2] in section 4.6.

Finally, for properly meromorphic  $f \in F_k$ , nothing in the construction changes except the topology. The punctured plane  $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  fails to be simply connected, the L-integral descends to homotopy classes of admissible contour pairs, and dragging a contour across an interior pole produces a jump given by a Cauchy residue weighted by the kernel carried by the crossing cycle —  $\tau_p^{s-1}$  for the  $S$ -segment,  $k_T(\tau_p, s)$  for the  $T$ -segment. Meromorphicity is manifest rather than accommodated: interior poles cannot affect finiteness, only topology, and the functional answers topology with computable wall-crossing residues.

**Theorem 1.7** (*L*-function for meromorphic modular forms). *Let  $k \in 2\mathbb{Z}$  and let  $f$  be a meromorphic modular form of weight  $k$  on  $\Gamma$ . Write  $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  for the upper half-plane with the  $\Gamma$ -orbit of the interior poles of  $f$  removed, and call the pair  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  admissible if  $\tau_0 \in \mathbb{H}_f$  and both contours lie in  $\mathbb{H}_f$ , with  $\gamma_{\tau_0}^S$  running from  $S^{-1}\tau_0 = -1/\tau_0$  to  $\tau_0$  and  $\gamma_{\tau_0}^T$  from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$ . Define  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  by (1.10) with the Hurwitz-zeta kernel  $\tilde{k}_T(\tau, s)$  of (1.7). Then:*

- (a) *Both integrals in (1.10) are finite for every admissible  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and every  $s \in \mathbb{C}$ .*
- (b)  *$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  depends on the admissible pair only through its homotopy class in  $\mathbb{H}_f$ : any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint  $\tau_0$  — leaves the value unchanged.*
- (c) *If two admissible configurations  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and  $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$  lie in homotopy classes related by a single crossing of an interior pole  $\tau_p \in \mathbb{H}$  of  $f$ , then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = \mathcal{R}_{\tau_p}(f, s), \quad (1.12)$$

where for a simple pole  $\text{Res}(f, \tau_p)$ ,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \left[ \sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (1.13)$$

with  $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$  the winding-number contributions of  $\gamma_{\tau_0}^S$  and  $\gamma_{\tau_0}^T$  across  $\tau_p$  (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in  $\tau_p$  of (1.13); the closed form is given in Section 5.

Restricted to meromorphic forms regular at the cusp, the  $L$ -integral reproduces the deformed-Mellin  $L$ -function of [3]; the precise compatibility statement is Corollary ?? in Section 5.

In sum: a single  $T$ -normalized cocycle value, read at integer  $s$ , is the period polynomial (Theorem 1.4); read at general  $s$ , it is the completed  $L$ -function (Theorem 1.6); read on nontrivial homotopy classes, it is the  $L$ -function of a meromorphic modular form with its wall-crossing structure (Theorem 1.7).

The remainder of this note is organized as follows. Section 2 introduces the period-defining cocycle and its moment expansion in the polynomial coefficient system  $V_{k-2}$ , making the narrative above precise. Section 3 solves the  $T$ -normalizing coboundary equation in closed Bernoulli form, identifies the resulting two-segment kernels, and proves Theorem 1.4. Section 4 derives the Hurwitz-zeta kernel  $\tilde{k}_T$  as the all- $s$  solution of the same coboundary equation, proves Theorem 1.6 via a Hurwitz-unfolding lemma, and contains an explicit worked example at  $k = 12$  exhibiting the Bernoulli-polynomial collapse of  $\tilde{k}_T$  at integer  $s$  together with numerical values of  $\omega^\pm(\Delta)$  and  $\eta^\pm(\hat{\Delta})$ . Section 5 proves Theorem 1.7 and Corollary ?. Section 6 collects conceptual framings (holonomy/Wilson-loop interpretation, on-shell-amplitudes analogy, the regularisation-as-symptom heuristic, the open question on the geometric content of the  $\tau^{s-1} f d\tau$  Mellin kernel, and pointers to the sequel on systematic per-mode theory for interior-pole forms). Appendix B records an alternative derivation of the Hurwitz-zeta kernel from contour-invariance of the two-segment functional: starting from the classical Mellin transform split at the contour endpoints  $\{-1/\tau_0, \tau_0\}$  and demanding that the resulting incomplete-gamma tail be representable as a  $T$ -segment integral, one derives a shift-difference equation whose solution is the Hurwitz-zeta combination of (1.7).

## 2 The period-defining cocycle and its moment expansion

**The polynomial coefficient system.** Let  $V_n := \mathbb{C}[X, Y]_n$  be the space of complex homogeneous polynomials in two auxiliary variables  $X, Y$  of degree  $n$ , equipped with the right  $\Gamma$ -action

$$(P|_\gamma)(X, Y) = P(aX + bY, cX + dY) \quad \text{for } \gamma = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \Gamma. \quad (2.1)$$

For a meromorphic modular form  $f$  of weight  $k$ , write  $n := k - 2$  throughout and form the  $V_n$ -valued 1-form on  $\mathbb{H}$

$$\omega_f(\tau) := f(\tau) (X - \tau Y)^n d\tau. \quad (2.2)$$

Modularity of  $f$  implies that  $\omega_f$  is  $\Gamma$ -equivariant in the standard sense [12]: under  $\tau \mapsto \gamma^{-1}\tau$ , the weight- $k$  transformation of  $f$  cancels the weight- $(-n)$  slash on  $(X - \tau Y)^n$ , so the  $V_n$ -valued 1-form  $\omega_f$  is invariant under the simultaneous action of  $\Gamma$  on  $\tau$  and on the auxiliary variables  $(X, Y)$ .

The construction of  $\omega_f$  above uses only the weight- $k$  modularity of  $f$  and makes no cuspidality assumption:  $\omega_f$  is well-defined for every meromorphic modular form  $f \in M_k^1$  (and indeed for meromorphic forms with poles in the fundamental domain, provided contours avoid the  $\Gamma$ -orbit of those poles), with the cuspidal hypothesis  $f \in S_k^1$  entering only later via Lemma 3.1(c) as the solvability condition for the period-polynomial coboundary equation.

**The cocycle.** The two-segment cocycle construction used in the present note is the standard Eichler–Shimura cohomological setup [12], in which a weakly holomorphic modular form is integrated against the polynomial coefficient system  $(X - \tau Y)^n$  along contours from  $\gamma^{-1}\tau_0$  to  $\tau_0$  for the generators  $\gamma \in \{S, T\}$  of  $\Gamma$ , with periods and quasi-periods extracted from the resulting 1-cocycle modulo coboundaries. The author was led to this presentation, and to its quasi-period reformulation, through personal correspondence with F. Brown [5]. Concretely: fix a basepoint  $\tau_0 \in \mathbb{H}$  not on the  $\Gamma$ -orbit of any pole of  $f$ , and define

$$C_\gamma^f(\tau_0) := (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} \omega_f = (2\pi i)^{n+1} \int_{\gamma^{-1}\tau_0}^{\tau_0} f(\tau) (X - \tau Y)^n d\tau \in V_n \otimes \mathbb{C}, \quad (2.3)$$

the contour being any path from  $\gamma^{-1}\tau_0$  to  $\tau_0$  in  $\mathbb{H}$  avoiding the  $\Gamma$ -orbit of the poles of  $f$ . The integral is independent of the choice of path within the same homotopy class in  $\mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$ .

**Lemma 2.1** (Cocycle property). *For all  $g, h \in \Gamma$ ,*

$$C_{gh}^f(\tau_0) = C_g^f(\tau_0)|_h + C_h^f(\tau_0). \quad (2.4)$$

*Equivalently, the map  $\gamma \mapsto C_\gamma^f(\tau_0)$  is a 1-cocycle of  $\Gamma$  with coefficients in  $V_n \otimes \mathbb{C}$ .*

*Proof.* Using  $(gh)^{-1} = h^{-1}g^{-1}$  and path-additivity through the intermediate point  $h^{-1}\tau_0$ ,

$$\int_{(gh)^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{\tau_0} = \int_{h^{-1}g^{-1}\tau_0}^{h^{-1}\tau_0} + \int_{h^{-1}\tau_0}^{\tau_0}. \quad (2.5)$$

In the first integral on the right substitute  $\tau = h^{-1}\sigma$ , so  $\sigma$  runs from  $g^{-1}\tau_0$  to  $\tau_0$ . The  $\Gamma$ -equivariance of  $\omega_f$  converts the integrand to  $\omega_f(\sigma)|_h$ , yielding the first summand of (2.4). The second integral gives the second summand directly.  $\square$

A cocycle on  $\Gamma = \mathrm{SL}_2(\mathbb{Z})$  is determined by its values on the two generators

$$S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad (2.6)$$

subject to the relations  $S^2 = (ST)^3 = -I$  (which act trivially on  $V_n$  for  $n$  even). We write

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \int_{-1/\tau_0}^{\tau_0} \omega_f, \quad C_T^f(\tau_0) = (2\pi i)^{n+1} \int_{\tau_0-1}^{\tau_0} \omega_f, \quad (2.7)$$

since  $S^{-1}\tau_0 = -1/\tau_0$  and  $T^{-1}\tau_0 = \tau_0 - 1$ . These are the two integrals over the contour pair (1.5).

**Moment expansion.** Expanding  $(X - \tau Y)^n = \sum_{\ell=0}^n \binom{n}{\ell} (-\tau)^\ell X^{n-\ell} Y^\ell$  and pulling the polynomial out of the integral defines the moment integrals

$$I_S^\ell(f, \tau_0) := \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad I_T^\ell(f, \tau_0) := \int_{\tau_0-1}^{\tau_0} f(\tau) \tau^\ell d\tau, \quad (2.8)$$

for  $\ell = 0, 1, \dots, n$ , in terms of which

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell, \quad (2.9)$$

and similarly for  $C_T^f(\tau_0)$ . Theorem 1.4 extracts  $\omega^\pm(f)$  from particular linear combinations of these moments.

**Basepoint dependence.** Varying  $\tau_0$  changes  $C^f$  by a coboundary, provided  $f$  has no interior poles. Define an antiderivative  $F(\tau)$  of  $\omega_f$  (so  $dF = \omega_f$ ) on a simply connected open subset of  $\mathbb{H}$  avoiding  $\Gamma \cdot f^{-1}(\infty)$ . Then for any two basepoints  $\tau_0, \tau'_0$  in the same connected component, path-additivity and the  $\Gamma$ -equivariance of  $\omega_f$  give

$$C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = (P|_\gamma - P) \quad \text{with} \quad P = (2\pi i)^{n+1} (F(\tau_0) - F(\tau'_0)) = (2\pi i)^{n+1} \int_{\tau'_0}^{\tau_0} \omega_f \in V_n \otimes \mathbb{C}. \quad (2.10)$$

This is the coboundary  $\delta P(\gamma) := P|_\gamma - P$ . When  $f$  has interior poles, dragging  $\tau_0$  across a pole picks up a Cauchy residue contribution beyond (2.10); this wall-crossing structure is the content of Theorem 1.7 in Section 5.

**Parabolic cohomology and periods.** The cohomology class  $[C^f] \in H^1(\Gamma; V_n)$  is independent of  $\tau_0$  (on the connected component) and of the choice of  $\delta P$  representative. Its restriction to the cusp stabilizer  $\langle T \rangle \cong \mathbb{Z}$  lands in  $H^1(\langle T \rangle; V_n) = V_n / \mathrm{Im} \delta_T \cong \mathbb{C} \cdot [X^n]$ , and by Lemma 3.1(c) the restricted class is  $(2\pi i)^{n+1} c_f(0) \cdot [X^n]$ : the residue of  $\omega_f$  at the cusp, valued in the  $T$ -coinvariants. The class is *parabolic* — cohomologous to a cocycle with vanishing  $T$ -value — if and only if this residue vanishes, i.e. if and only if  $f \in S_k^!$ . The classical Eichler–Shimura theorem [12] identifies  $H^1(\Gamma; V_n) \cong S_k \oplus \overline{S_k} \oplus \mathbb{C}_{\mathrm{Eis}}$  and the parabolic subspace  $H_{\mathrm{par}}^1(\Gamma; V_n) \cong S_k \oplus \overline{S_k}$ ; on cocycles normalized so that  $c_T = 0$ , the assignment  $c \mapsto c_S$  further identifies  $H_{\mathrm{par}}^1(\Gamma; V_n) \cong W_{k-2}/\mathbb{C}(X^n - Y^n)$ , with the quotient line accounting for the residual freedom  $P \mapsto P + \mathbb{C}Y^n$ . When  $\dim S_k = 1$ , each parity eigenspace of  $H_{\mathrm{par}}^1$  is one-dimensional, and the two coordinates of a parabolic class in Brown's rational basis  $p_f^\pm$  are its periods:  $[C^f]^\pm = \omega^\pm(f)[p_f^\pm]$  for  $f \in S_k$ , while the quasi-periods  $\eta^\pm$  arise as the coordinates of the class  $[C^{f'}]$  of a weakly holomorphic Hecke partner  $f' \in S_k^!$  (at  $k = 12$ :  $f = \Delta$ ,  $f' = \widehat{\Delta}$ ) [1].

### 3 Period extraction: proof of Theorem 1.4

Set  $n := k - 2$  throughout. The argument is  $k$ -uniform on the  $\dim S_k = 1$  family  $k \in \{12, 16, 18, 20, 22, 26\}$ : every step below — the cocycle structure of Section 2, the  $\delta_T$ -solvability of Lemma 3.1, the Bernoulli closed-form inversion of Lemma 3.2, the Hurwitz-zeta kernel identification of Proposition 3.6, and the  $\ell$ - and  $\tau_0$ -independence of Corollary 3.8 — holds for every even  $k$  with the Brown cuspidal-basis coefficient  $c_S^{\ell-1}$  replaced by the relevant weight's basis-polynomial coefficient. The lower even weights  $k \in \{2, 4, 6, 8, 10, 14\}$  in the  $\dim S_k \leq 1$  family have  $S_k = 0$ , so the theorem is vacuous there. Numerical verification at  $k = 12$  against Brown's published values for the canonical pair  $f \in \{\Delta, \hat{\Delta}\}$  is consolidated in Appendix A.

**Lemma 3.1** (Solvability of  $\delta_T P = C_T^f$  is cuspidality of  $f$ ). *Define the linear endomorphism  $\delta_T: V_n \rightarrow V_n$  by  $\delta_T(P) := P|_T - P$ , where  $T \in \Gamma$  acts on  $V_n$  as  $(X, Y) \mapsto (X + Y, Y)$ . Then:*

- (a)  $\ker(\delta_T) = \mathbb{C} \cdot Y^n$ .
- (b)  $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$ .
- (c) For any meromorphic modular form  $f$  of weight  $k = n + 2$  and any  $\tau_0 \in \mathbb{H}$  admissible for  $f$ ,

$$[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} \cdot c_f(0), \quad (3.1)$$

where  $c_f(0)$  is the constant Fourier coefficient of  $f$ .

Consequently, the equation  $\delta_T P = C_T^f(\tau_0)$  is solvable for  $P \in V_n$  if and only if  $c_f(0) = 0$ , i.e. if and only if  $f$  is a (weak) cusp form. When solvable,  $P$  is unique modulo  $\mathbb{C} \cdot Y^n$ .

*Proof.* (a)  $T$  acts on  $V_n$  as the translation  $X \mapsto X + Y$ , fixing only powers of  $Y$ . Hence  $T$ -invariants in  $V_n$ , equivalently  $\ker(\delta_T)$ , are exactly  $\mathbb{C} \cdot Y^n$ .

(b) Writing  $P = \sum_{\ell=0}^n p_\ell X^{n-\ell} Y^\ell$  and expanding  $(X + Y)^{n-\ell}$  binomially,

$$[\delta_T P]_{X^{n-m} Y^m} = \sum_{\ell=0}^{m-1} \binom{n-\ell}{m-\ell} p_\ell \quad (m = 0, 1, \dots, n). \quad (3.2)$$

This is strictly lower-triangular in the ordered basis  $(X^n, X^{n-1}Y, \dots, Y^n)$ : the  $m = 0$  row is the empty sum, identically zero, and the off-diagonal entries  $\binom{n-\ell}{m-\ell}$  for  $\ell < m$  are positive. The  $m = 0$  row forces  $[\delta_T P]_{X^n} = 0$  for every  $P$ , while rows  $m = 1, \dots, n$  determine  $p_0, \dots, p_{n-1}$  uniquely from any prescribed right-hand side, with  $p_n$  unconstrained. Hence  $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$ , and the solution  $P$  is unique modulo  $\mathbb{C} \cdot Y^n$ .

(c) The moment expansion (2.7) gives  $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} I_T^0(f, \tau_0)$  where  $I_T^0(f, \tau_0) = \int_{\tau_0-1}^{\tau_0} f(\tau) d\tau$ . Substituting the Fourier expansion  $f(\tau) = \sum_n c_f(n) e^{2\pi i n \tau}$ , absolutely convergent on the compact  $T$ -segment, and integrating term by term,

$$\int_{\tau_0-1}^{\tau_0} e^{2\pi i n \tau} d\tau = \begin{cases} 1 & \text{if } n = 0, \\ 0 & \text{if } n \neq 0, \end{cases} \quad (3.3)$$

by  $T$ -periodicity over a unit segment. Hence  $I_T^0(f, \tau_0) = c_f(0)$  for every admissible  $\tau_0$ , which proves (3.1).

Combining (b) and (c) gives the solvability statement.  $\square$

**Lemma 3.2** (Bernoulli closed form for  $\delta_T^{-1}$ ). *For every  $Q \in V_n$  with vanishing  $X^n$ -coefficient there exists  $P \in V_n$  unique modulo  $\mathbb{C} \cdot Y^n$  with  $\delta_T P = Q$ . Writing  $Q = \sum_{r=1}^n q_r X^{n-r} Y^r$  and  $P = \sum_{m=0}^n p_m X^{n-m} Y^m$ , the solution is given in closed form by*

$$p_m = \sum_{r=1}^{m+1} \frac{B_{m-r+1} (n-r)!}{(m-r+1)! (n-m)!} q_r \quad (m = 0, 1, \dots, n-1), \quad (3.4)$$

where  $B_\ell$  are Bernoulli numbers ( $B_0 = 1$ ,  $B_1 = -\frac{1}{2}$ ,  $B_2 = \frac{1}{6}$ ,  $B_{2\ell+1} = 0$  for  $\ell \geq 1$ ); the coefficient  $p_n$  is unconstrained and parametrises  $\ker(\delta_T) = \mathbb{C} \cdot Y^n$ .

*Proof.* The operator identity  $T = e^{Y\partial_X}$  on  $V_n$  (since  $T$  acts as the translation  $X \mapsto X + Y$  by  $Y$ ) gives  $\delta_T = e^{Y\partial_X} - 1$ . The operator  $Y\partial_X$  is nilpotent on  $V_n$ : each application reduces the  $X$ -degree by one, so  $(Y\partial_X)^{n+1} = 0$  on  $V_n$ . Consequently, the formal power series in  $Y\partial_X$  obtained from the Bernoulli generating function  $z/(e^z - 1) = \sum_{k \geq 0} (B_k/k!) z^k$  truncates to a finite  $\mathbb{Q}$ -linear combination of operators on  $V_n$ , and supplies the closed-form inverse

$$\delta_T^{-1} = (Y\partial_X)^{-1} \frac{Y\partial_X}{e^{Y\partial_X} - 1} = \sum_{k \geq 0} \frac{B_k}{k!} (Y\partial_X)^{k-1}, \quad (3.5)$$

in which the  $k = 0$  summand is the formal antiderivative  $(Y\partial_X)^{-1}$ , acting on monomials by  $(Y\partial_X)^{-1}(X^a Y^b) = X^{a+1} Y^b / (a+1)$  (for  $b \geq 1$ ), and the  $k \geq 1$  summands are differential operators that eventually annihilate any polynomial of bounded  $X$ -degree. Applied to  $X^{n-r} Y^r$  for  $r \geq 1$ ,

$$\delta_T^{-1}(X^{n-r} Y^r) = \sum_{k=0}^{n-r+1} \frac{B_k}{k!} \frac{(n-r)!}{(n-r-k+1)!} X^{n-r-k+1} Y^{r+k-1}. \quad (3.6)$$

Summing  $Q = \sum_r q_r X^{n-r} Y^r$  termwise and reading off the  $X^{n-m} Y^m$  coefficient identifies  $k = m-r+1$  (so  $r$  runs over  $\{1, \dots, m+1\}$ ,  $k$  over  $\{0, \dots, m\}$ ), yielding (3.4). Existence and uniqueness modulo  $\mathbb{C} \cdot Y^n$  follow from Lemma 3.1(a, b).  $\square$

**Definition 3.3** (Bernoulli  $T$ -segment kernel  $\widehat{k}_T(\ell, \tau)$ ). *For non-negative integers  $l$  with  $l \leq n-1$ , set the  $\mathbb{Q}$ -rational coefficients*

$$\beta_{l,r} := \frac{(-1)^r \binom{n}{r} B_{l-r+1} (n-r)!}{(l-r+1)! (n-l)!} \quad \text{for } 1 \leq r \leq l+1, \quad (3.7)$$

and  $\beta_{l,r} := 0$  for  $r$  outside  $\{1, \dots, l+1\}$ . For each interior  $\ell \in \{2, \dots, k-2\}$ , define the  $\mathbb{Q}$ -rational polynomial

$$\widehat{k}_T(\ell, \tau) := \sum_{r=1}^n \left[ \beta_{\ell-1,r} - (-1)^{\ell-1} \beta_{n-\ell+1,r} \right] \tau^r \in \mathbb{Q}[\tau], \quad (3.8)$$

of degree  $\leq n$ .

**Lemma 3.4** (Integer- $s$  collapse of the Hurwitz-zeta  $T$ -kernel). *For each integer  $\ell \in \{2, \dots, k-2\}$ , the Hurwitz-zeta  $T$ -kernel  $\widetilde{k}_T(\tau, s)$  of (1.7) specialised to  $s = \ell$  collapses to the  $\mathbb{Q}$ -rational polynomial*

$$\widetilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (3.9)$$

of degree at most  $k-2$ .

*Proof.* The Hurwitz-zeta collapse  $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$  for non-negative integer  $m$ , applied at  $m = \ell - 1$  and  $m = k - \ell - 1$  (both non-negative for  $\ell \in \{2, \dots, k-2\}$ ), together with  $e^{i\pi(\ell-1)} = (-1)^{\ell-1}$ , converts (1.7) at  $s = \ell$  to (3.9).  $\square$

**Lemma 3.5** (Bernoulli-form vs. Hurwitz-form  $T$ -kernel: polynomial identity and integral equivalence). *For each interior  $\ell \in \{2, \dots, k-2\}$ , the Bernoulli  $T$ -kernel  $\widehat{k}_T(\ell, \tau)$  of (3.8) and the Hurwitz  $T$ -kernel  $\widetilde{k}_T(\tau, \ell)$  of (3.9) are related by the polynomial identity*

$$\widehat{k}_T(\ell, \tau) = (-1)^{\ell-1} \binom{n}{\ell-1} \left[ \widetilde{k}_T(\tau, \ell) - \widetilde{k}_T(0, \ell) \right] \in \mathbb{Q}[\tau]. \quad (3.10)$$

*Equivalently, the difference  $\widehat{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(0, \ell) \in \mathbb{Q}$  is a constant. Consequently, for every  $f \in S_k^!$  and every  $\tau_0 \in \mathbb{H}$ ,*

$$\int_{\gamma_{\tau_0}^T} f(\tau) \widehat{k}_T(\ell, \tau) d\tau = (-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau. \quad (3.11)$$

*Proof. Polynomial identity.* Both sides of (3.10) are  $\mathbb{Q}$ -rational polynomials in  $\tau$  of degree  $\leq k-2$ , and both have vanishing constant term: the left side by construction ((3.8) sums from  $r = 1$ ), and the right side because  $\widetilde{k}_T(\tau, \ell) - \widetilde{k}_T(0, \ell)$  vanishes at  $\tau = 0$ . By direct comparison of the  $\tau^r$  coefficients for  $1 \leq r \leq k-2$  using (3.8) for the left side and (3.9) for the right side, together with the standard Bernoulli polynomial expansion  $B_n(\tau+1) = \sum_{s=0}^n \binom{n}{s} B_s \tau^{n-s} + n \tau^{n-1}$ , the two polynomial coefficients agree. Hence the polynomial identity (3.10) holds in  $\mathbb{Q}[\tau]$ .

*Integral equivalence.* The difference  $\widehat{k}_T(\ell, \tau) - (-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(\tau, \ell) = -(-1)^{\ell-1} \binom{n}{\ell-1} \widetilde{k}_T(0, \ell) \in \mathbb{Q}$  is constant by (3.10). For  $f \in S_k^!$  the integral of any constant over the  $T$ -segment vanishes by cuspidality:  $\int_{\gamma_{\tau_0}^T} f(\tau) d\tau = c_f(0) = 0$  (Lemma 3.1(c)). Hence the constant difference contributes zero and (3.11) follows.  $\square$

**Proposition 3.6** (Integral representation of  $r_f$  at the interior monomial). *Let  $f \in S_k^!$  and  $\tau_0 \in \mathbb{H}$ . Let  $P \in V_n/\text{CY}^n$  be the unique solution of  $\delta_T P = (2\pi i)^{-(n+1)} C_T^f(\tau_0)$  from Lemma 3.1 and Lemma 3.2, and define the modified-cocycle  $S$ -value*

$$r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P) \in V_n \otimes \mathbb{C}. \quad (3.12)$$

*For each interior  $\ell \in \{2, \dots, k-2\}$ , the coefficient of the monomial  $X^{n-(\ell-1)} Y^{\ell-1}$  in  $r_f$  admits the integral representation*

$$[r_f]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} (-1)^{\ell-1} \binom{n}{\ell-1} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau \right], \quad (3.13)$$

*with  $\widetilde{k}_T(\tau, s)$  the Hurwitz-zeta kernel of (1.7).*

*Proof.  $S$ -piece.* The moment expansion (2.7) gives

$$C_S^f(\tau_0) = (2\pi i)^{n+1} \sum_{\ell=0}^n \binom{n}{\ell} (-1)^\ell I_S^\ell(f, \tau_0) X^{n-\ell} Y^\ell, \quad (3.14)$$

so  $[C_S^f(\tau_0)]_{X^{n-(\ell-1)} Y^{\ell-1}} = (2\pi i)^{n+1} \binom{n}{\ell-1} (-1)^{\ell-1} I_S^{\ell-1}(f, \tau_0)$ , with  $I_S^{\ell-1} = \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau$ ; this contributes the  $S$ -segment piece of (3.13) against the bare Mellin kernel  $\tau^{\ell-1}$ .

*T-piece, Bernoulli form.*  $S$  acts on  $V_n$  as  $(X, Y) \mapsto (-Y, X)$ , so for  $P = \sum_l p_l X^{n-l} Y^l$  one has  $[P|_{S-P}]_{X^{n-(\ell-1)}Y^{\ell-1}} = (-1)^{\ell-1} p_{n-\ell+1} - p_{\ell-1}$  ( $n$  even). By Lemma 3.2, each  $p_l$  ( $0 \leq l \leq n-1$ ) is the explicit  $\mathbb{Q}$ -linear combination  $p_l = \sum_{r=1}^{l+1} \beta_{l,r} I_T^r(f, \tau_0)$  with  $\beta_{l,r}$  from (3.7) of Definition 3.3, obtained by substituting the rescaled  $T$ -moments  $q_r = \binom{n}{r} (-1)^r I_T^r$  of  $(2\pi i)^{-(n+1)} C_T^f(\tau_0)$  into (3.4). In terms of  $\widehat{k}_T(\ell, \tau)$  of (3.8), the  $T$ -segment piece of  $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}}$  reads  $(2\pi i)^{n+1} \int_{\gamma_{\tau_0}^T} f(\tau) \widehat{k}_T(\ell, \tau) d\tau$ .

*T-piece, Hurwitz form.* By Lemma 3.5, (3.11), the Bernoulli  $T$ -integral of the preceding paragraph equals  $(-1)^{\ell-1} \binom{n}{\ell-1} \int_{\gamma_{\tau_0}^T} f(\tau) \widehat{k}_T(\tau, \ell) d\tau$ . Combined with the  $S$ -piece, this yields (3.13) with the bare Hurwitz kernel  $\widetilde{k}_T(\tau, \ell)$  inside.  $\square$

**Lemma 3.7** (Brown-basis extraction of  $\omega^\pm(f)$ ). *Let  $r_f \in V_n \otimes \mathbb{C}$  be the modified-cocycle  $S$ -value of (3.12). The Eichler–Shimura / Brown cuspidal-basis decomposition ([1, 12]) reads*

$$r_f = \omega^+(f) p_f^+ + \omega^-(f) p_f^- + (\text{Eisenstein-direction term}), \quad (3.15)$$

*with the Eisenstein-direction term contributing only to the boundary monomials  $X^n, Y^n$  of  $r_f$  (as it is a multiple of the coboundary  $Y^n|_S - Y^n = X^n - Y^n$ ). Combined with the integral representation (3.13) of Proposition 3.6, this yields*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau \right] \quad (3.16)$$

*for every interior  $\ell \in \{2, \dots, k-2\}$ , with parity  $\pm$  determined by  $\ell-1 \bmod 2$  and Brown basis coefficient  $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$ . Identity (3.16) is identity (1.9) of Theorem 1.4.*

*Proof.* For interior  $\ell-1 \in \{1, \dots, n-1\}$ , the Eisenstein-direction term in (3.15) contributes zero at the monomial  $X^{n-(\ell-1)}Y^{\ell-1}$ , so  $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}} = \omega^\pm(f) c_S^{\ell-1}$ . Substituting (3.13) and dividing through by  $c_S^{\ell-1}$  yields (3.16).  $\square$

**Corollary 3.8** (Basepoint independence). *The two-segment linear functional of Theorem 1.4,  $\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} [\int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \widetilde{k}_T(\tau, \ell) d\tau]$  of (1.9), is independent of  $\tau_0 \in \mathbb{H}$ , and returns the same  $\omega^\pm(f)$  for every interior  $\ell \in \{2, \dots, k-2\}$  in a fixed parity sector.*

*Proof.*  $\tau_0$ -independence. By (3.12), namely  $r_f := C_S^f(\tau_0) - (2\pi i)^{n+1} (P|_S - P)$ ,  $r_f$  depends on  $\tau_0$  only through the cohomology class  $[C^f] \in H^1(\Gamma; V_n \otimes \mathbb{C})$  and the choice of coboundary representative  $P$ . The class is  $\tau_0$ -invariant by the basepoint-shift coboundary identity  $C_\gamma^f(\tau'_0) - C_\gamma^f(\tau_0) = P|_\gamma - P$  for the antiderivative  $P = (2\pi i)^{n+1} (F(\tau_0) - F(\tau'_0))$  of (2.10) (valid for  $f$  without interior poles), and the coboundary representative  $P(\tau_0)$  from Lemma 3.2 is the unique-modulo- $\mathbb{C}Y^n$  solution of  $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$ . The residual  $\mathbb{C}Y^n$  ambiguity lies in the Eisenstein direction:  $Y^n|_S - Y^n = X^n - Y^n$  is a coboundary that contributes only to the boundary monomials  $X^n, Y^n$  of  $r_f$ . Therefore the cuspidal coordinates of  $r_f$  — the  $\omega^\pm(f)$  — are  $\tau_0$ -invariant.

$\ell$ -independence. This carries no content beyond Lemma 3.7, where  $r_f$  is shown to lie in the 1-dimensional cuspidal line  $\mathbb{C} p_f^\pm$ : reading any interior monomial of  $r_f$  and dividing by  $c_S^{\ell-1} = [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}}$  returns the same  $\omega^\pm(f)$ , for every interior  $\ell$  in the parity sector.  $\square$

With Proposition 3.6 (integral representation of  $[r_f]$  at interior monomials), Lemma 3.4 (integer- $s$  collapse of  $\widetilde{k}_T$ ), Lemma 3.7 (Brown-basis extraction of  $\omega^\pm$ ), and Corollary 3.8 ( $\ell$ - and  $\tau_0$ -independence) in hand, we assemble the proof of Theorem 1.4. For reference, the theorem statement is:

**Theorem 3.9** (Periods of  $f \in S_k^!$  as an explicit linear functional ( $\dim S_k = 1$ ) – Theorem 1.4, restated). *Let  $k \in 2\mathbb{Z}$  with  $\dim S_k \leq 1$ , and let  $f \in S_k^!$ . For the substantive cases  $k \in \{12, 16, 18, 20, 22, 26\}$  where  $\dim S_k = 1$ , the period and quasi-period coefficients  $\omega^\pm(f)$  are given by the explicit two-segment linear functional*

$$\omega^\pm(f) = \frac{(2\pi i)^{k-1} (-1)^{\ell-1} \binom{k-2}{\ell-1}}{c_S^{\ell-1}} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right] \quad (3.17)$$

for every  $f \in S_k^!$ , every  $\tau_0 \in \mathbb{H}$ , and every interior  $\ell \in \{2, \dots, k-2\}$ , with parity  $\pm$  determined by  $\ell - 1 \bmod 2$ . Here  $c_S^{\ell-1} := [p_f^\pm]_{X^{n-(\ell-1)}Y^{\ell-1}} \in \mathbb{Q}$  is the Brown cuspidal-basis coefficient at the matched monomial ([1, 12]),  $\tilde{k}_T(\tau, s)$  is the Hurwitz-zeta  $T$ -kernel of (1.7), and its integer- $s = \ell$  specialisation collapses via  $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$  ( $m \geq 0$ ) to the  $\mathbb{Q}$ -rational polynomial

$$\tilde{k}_T(\tau, \ell) = -\frac{B_\ell(\tau+1)}{\ell} + (-1)^{\ell-1} \frac{B_{k-\ell}(\tau+1)}{k-\ell} \in \mathbb{Q}[\tau] \quad (3.18)$$

of degree at most  $k-2$ . The right side of (3.17) is independent of  $\tau_0 \in \mathbb{H}$ .

*Proof of Theorem 1.4, aka Theorem 3.9.* The proof proceeds in seven steps. Steps 1–3 motivate the operator  $\delta_T$  as the natural object whose inversion delivers the cuspidal cocycle representative. Step 4 (Lemma 3.2) performs the inversion in closed form. Steps 5–7 read  $\omega^\pm(f)$  off the resulting period polynomial.

**Step 1: the canonical cuspidal representative has  $C_T^f = 0$ .** Within the cohomology class  $[C^f] \in H^1(\Gamma; V_n)$ , consider the representative with vanishing  $T$ -component. When  $C_T = 0$ , the cocycle relations  $C_{S^2} = 0$  and  $C_{(ST)^3} = 0$  collapse onto  $C_S$  alone, and the surviving relations on  $C_S$  are exactly the two Eichler–Shimura period-polynomial relations  $r_f|(1+S) = 0$  and  $r_f|(1+U+U^2) = 0$  with  $U = TS$ . The  $C_T = 0$  representative therefore has its  $S$ -value in the Eichler–Shimura period-polynomial space, where Brown’s basis  $\{p_f^+, p_f^-, \text{Eisenstein}\}$  is defined and the cuspidal coordinates are  $\omega^\pm(f)$ . The proof below kills  $C_T^f$  by a coboundary modification.

**Step 2: kill  $C_T^f$  by a coboundary modification.** The only freedom inside the cohomology class is to add a coboundary  $\delta P(\gamma) := P|_\gamma - P$  for some  $P \in V_n$ . Under this modification,  $C_T^f(\tau_0) \mapsto C_T^f(\tau_0) - (P|_T - P)$ . Killing  $C_T^f$  becomes the equation

$$\delta_T P := P|_T - P = C_T^f(\tau_0), \quad (3.19)$$

to be solved for  $P \in V_n$ . This identifies the linear endomorphism  $\delta_T: V_n \rightarrow V_n$  of Lemma 3.1 as the precise object whose inversion delivers the desired coboundary representative.

**Step 3: solvability of  $\delta_T P = C_T^f$  is exactly cuspidality of  $f$ .** Lemma 3.1 computes  $[C_T^f(\tau_0)]_{X^n} = (2\pi i)^{n+1} c_f(0)$  and shows that  $\text{im}(\delta_T) = \{Q \in V_n : [Q]_{X^n} = 0\}$ . Solvability of (3.19) is therefore equivalent to  $c_f(0) = 0$ , i.e. to  $f$  being a (weak) cusp form. The cuspidality hypothesis  $f \in S_k^!$  is not a separate input to the construction: it is the solvability condition of the coboundary equation in Step 2.

**Step 4: closed-form Bernoulli inversion of  $\delta_T$ .** Lemma 3.2 inverts  $\delta_T$  in closed form, giving the coboundary representative  $P = \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n/\mathbb{C}Y^n$  explicitly in Bernoulli numbers.

**Step 5: the modified cocycle and its  $S$ -component  $r_f$ .** With  $P := \delta_T^{-1}((2\pi i)^{-(n+1)} C_T^f(\tau_0)) \in V_n/\mathbb{C}Y^n$  from Step 4, the modified cocycle  $C'^f := C^f - (2\pi i)^{n+1} \delta P$  has  $C'^f_T = 0$  by construction, and its  $S$ -value  $r_f := C'^f_S(\tau_0) = C^f_S(\tau_0) - (2\pi i)^{n+1} (P|_S - P)$  of (3.12) is the Eichler–Shimura period polynomial of  $f$ . By the Eichler–Shimura / Brown cuspidal-basis decomposition (3.15) of Lemma 3.7,  $\omega^\pm(f)$  are the explicit linear coordinates of  $r_f$  along  $p_f^\pm$  in the 1-dimensional cuspidal direction.

**Step 6: unpacking  $r_f$  into bare Mellin and Hurwitz integrals.** Substituting the moment expansion of  $C^f_S(\tau_0)$  from (2.7) and the Bernoulli expression for  $P$  from Step 4 into (3.12), and reading off the coefficient of  $X^{n-(\ell-1)}Y^{\ell-1}$  for each interior  $\ell \in \{2, \dots, k-2\}$ , Proposition 3.6 produces the integral representation (3.13) of  $[r_f]_{X^{n-(\ell-1)}Y^{\ell-1}}$  with the bare Mellin kernel  $\tau^{\ell-1}$  on the  $S$ -segment and the bare Hurwitz kernel  $\tilde{k}_T(\tau, \ell)$  on the  $T$ -segment. Lemma 3.4 (the integer- $s$  collapse of  $\tilde{k}_T$ ) exhibits the  $T$ -kernel as a  $\mathbb{Q}$ -rational polynomial of degree  $\leq k-2$ . Lemma 3.7 (Brown-basis extraction) then divides through by  $c_S^{\ell-1}$  to give (1.9).

**Step 7:  $\ell$ - and  $\tau_0$ -independence.** Corollary 3.8 supplies both:  $r_f$  is a single polynomial in the 1-dimensional cuspidal direction, so every interior monomial returns the same  $\omega^\pm(f)$  (parity-selected by  $\ell$ ), while  $\tau_0$ -invariance follows from  $[C^f]$  being a cohomology-class invariant (the residual  $\mathbb{C}Y^n$  ambiguity touching only the Eisenstein direction). □

**Corollary 3.10** (Incomplete Gamma Representation). *Let  $f \in S_k^!$ , where  $\dim(S_k) = 1$ , and let  $\tau_0 = i$ . The integral functional within period polynomial functional in Eq. (3.17) in Theorem 1.4 simplifies to*

$$i^{-s} \left( \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{\ell-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, \ell) d\tau \right) = \sum_{n \neq 0} c(n) \left( \frac{\Gamma(\ell, 2\pi n)}{(2\pi n)^\ell} + i^k \frac{\Gamma(k-\ell, 2\pi n)}{(2\pi n)^{k-\ell}} \right), \quad (3.20)$$

where  $c(n)$  are the  $q$ -series coefficients of  $f$ , i.e.  $f = \sum_n c(n) q^n$ .

*Proof.* We need to bring in the Hurwitz unfolding crapola here, I think, or something.

Steps are:

1. Note that at  $\tau_0 = i$ ,  $\gamma_{\tau_0}^S = 0$ , whereas  $\gamma_{\tau_0}^T$  goes simply between  $i-1$  and  $i$ .
2. Note that, because the  $q$ -series converges absolutely along the finite-contour, as  $\sim e^{-2\pi(n-C\sqrt{n})}$  for some constant  $C$ , we can exchange the  $q$ -series and the contour integral.
3. Recall that  $\tilde{k}_T(\tau, \ell) = \zeta_H(1-\ell, \tau+1) - e^{i\pi(\ell-1)} \zeta_H(1-(k-s), \tau+1)$
4. Show that  $\int_{i-1}^i d\tau e^{2\pi n \tau} \zeta_H(1-\ell, \tau+1) = \Gamma(\ell, 2\pi n)/(2\pi n)^\ell$  or whatever
5. Note that the same holds, with  $\ell \leftrightarrow k-\ell$  for the other term.<sup>1</sup>

Conclude the proof with combining these steps and giving the result. □

Explicitly, by using Eq. (3.20) at  $k = 12$  for the canonical pair  $f \in \{\Delta, \hat{\Delta}\}$ , where  $\Delta$  is the modular discriminant, and  $\hat{\Delta} := 1/q + \mathcal{O}(q^2) \in S_{12}^!$ , we reproduce the numerical, published values [1], in Table 1.

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<sup>1</sup>I am concerned that the relative factor of  $i^k$  won't be respected in this line of the proof. Is this a legit concern?

|               | $f = \Delta \in S_{12}$         | $f = \widehat{\Delta} \in S_{12}^!$ |
|---------------|---------------------------------|-------------------------------------|
| $\omega^+(f)$ | $-68\,916\,772.80959519\dots$   | $127\,202\,100\,647.17709\dots$     |
| $\omega^-(f)$ | $-i\,5\,585\,015.37931040\dots$ | $i\,10\,276\,732\,343.64913\dots$   |

Table 1: Numerical periods of  $\Delta$  and  $\widehat{\Delta}$  from [1], via pythonin incarnation of Eq. (3.20) in SOME FUNCTION IN [4].

## 4 The Hurwitz-zeta extension and the $L$ -function on $M_k^!$ : proof of Theorem 1.6

### 4.1 Overview and the origin of the kernel

We prove Theorem 1.6: the functional  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  of (4.10), built from the Hurwitz-zeta  $T$ -kernel  $\widetilde{k}_T(\tau, s)$  of (1.7), is finite and basepoint-independent on  $M_k^!$ , and at the elliptic basepoint  $\tau_0 = i$  collapses to the completed  $L$ -function in closed form. The argument has three movements: (i) the kernel obeys a shift-difference identity (Lemma 4.1) that renders the two-segment functional finite (Lemma 4.2) and locally constant in its contour data (Lemma 4.3), so  $L^*(f, s)$  is well defined; (ii) at  $\tau_0 = i$  the  $S$ -segment degenerates and a single per-mode *unfolding* (Proposition 4.4) turns the surviving  $T$ -integral into incomplete gammas; (iii) the resulting sum is the completed  $L$ -function. Lemmas 4.2 and 4.3 are stated for arbitrary meromorphic  $f$ , in preparation for Section 5; Theorem 1.6 is their specialisation to  $f \in M_k^!$ , where no interior pole obstructs the contour pair.

**Why a Hurwitz-zeta kernel.** The kernel (1.7) is the all- $s$  continuation of the integer- $s$  inversion of Section 3. There, killing the  $T$ -cocycle meant solving the shift-difference equation  $\delta_T P = P|_T - P = Q$  on  $V_n$ , and the operator identity  $\delta_T = e^{Y\partial_x} - 1$  inverted it through the Bernoulli generating function  $z/(e^z - 1) = \sum_k B_k z^k/k!$  (Lemma 3.2). Dropping the polynomial restriction, the kernel must solve the same shift-difference equation with a power-law right-hand side — the identity (4.1) of Lemma 4.1 below, whose right side is a combination of pure powers  $\tau^{\sigma-1}$ . The function that inverts the unit shift against a single power is the Hurwitz zeta,  $\zeta(\sigma, \tau + 1) - \zeta(\sigma, \tau) = -\tau^{-\sigma}$ , so a combination of Hurwitz zetas is the natural solution, and the Bernoulli inversion of Section 3 is exactly its integer- $s$  shadow through  $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$  — the Bernoulli numbers being zeta values at non-positive integers. An alternative derivation, starting from the classical Mellin transform and demanding contour-invariance, is recorded in Appendix B; the present section is self-contained without it.

### 4.2 Well-definedness of the two-segment functional

Write  $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  for the upper half-plane with the  $\Gamma$ -orbit of the interior poles of  $f$  removed, and call a contour pair  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  *admissible* if  $\tau_0 \in \mathbb{H}_f$  and both segments lie in  $\mathbb{H}_f$ . For  $f \in M_k^!$  there are no interior poles, so  $\mathbb{H}_f = \mathbb{H}$  and every pair is admissible.

**Lemma 4.1** (Shift-difference identity for  $\widetilde{k}_T$ ). *The kernel  $\widetilde{k}_T(\tau, s)$  of (1.7) is analytic in  $\tau \in \mathbb{H}$  and meromorphic in  $s \in \mathbb{C}$ , and satisfies*

$$\widetilde{k}_T(\tau, s) - \widetilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)} \tau^{k-1-s} \quad (4.1)$$

for every  $\tau \in \mathbb{H}$  and every  $s \in \mathbb{C}$ .

*Proof.* Analyticity in  $\tau$  is immediate: each Hurwitz factor  $\zeta(\cdot, \tau + 1)$  has second argument with  $\operatorname{Re}(\tau + 1) > 1$  on  $\mathbb{H}$ , where  $\zeta(\sigma, a)$  is analytic in  $a$ . For the identity, the Hurwitz raising relation  $\zeta(\sigma, a + 1) - \zeta(\sigma, a) = -a^{-\sigma}$  [6, §25.11.3] at  $(\sigma, a) = (1 - s, \tau)$  and at  $(\sigma, a) = (s - (k - 1), \tau)$  gives

$$\zeta(1 - s, \tau + 1) - \zeta(1 - s, \tau) = -\tau^{s-1}, \quad \zeta(s - (k - 1), \tau + 1) - \zeta(s - (k - 1), \tau) = -\tau^{k-1-s}. \quad (4.2)$$

Since  $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)}\zeta(s - (k - 1), \tau + 1)$  and, shifting  $\tau \mapsto \tau - 1$ ,  $\tilde{k}_T(\tau - 1, s) = \zeta(1 - s, \tau) - e^{i\pi(s-1)}\zeta(s - (k - 1), \tau)$ , subtracting gives  $\tilde{k}_T(\tau, s) - \tilde{k}_T(\tau - 1, s) = -\tau^{s-1} + e^{i\pi(s-1)}\tau^{k-1-s}$ , which is (4.1).  $\square$

**Lemma 4.2** (Finiteness). *Let  $f$  be any meromorphic modular form of weight  $k$  on  $\Gamma$ . For every admissible contour pair  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and every  $s \in \mathbb{C}$ , both integrals in (4.10) are finite, so  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is well defined. In particular this holds for every  $f \in M_k^!$  and every  $\tau_0 \in \mathbb{H}$ .*

*Proof.* By admissibility both segments are compact and lie in  $\mathbb{H}_f$ , on which  $f$  is holomorphic; the kernels  $\tau^{s-1}$  and  $\tilde{k}_T(\tau, s)$  are analytic in  $\tau \in \mathbb{H}$  for each fixed  $s$  (Lemma 4.1). A continuous integrand on a compact contour has finite integral. For  $f \in M_k^!$  there are no interior poles,  $\mathbb{H}_f = \mathbb{H}$ , and every  $\tau_0 \in \mathbb{H}$  yields an admissible pair.  $\square$

**Lemma 4.3** (Local constancy and winding-invariance). *Let  $f$  be any meromorphic modular form of weight  $k$  on  $\Gamma$ . Then  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is unchanged under any continuous deformation of the admissible data  $(\tau_0, \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  that does not move a segment across a point of  $\Gamma \cdot f^{-1}(\infty)$ ; equivalently, it depends on the admissible pair only through the winding of the two segments about the interior poles of  $f$ . In particular, for  $f \in M_k^!$  — where  $\mathbb{H}_f = \mathbb{H}$  is simply connected —  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0 \in \mathbb{H}$  and of the contour pair for every  $s \in \mathbb{C}$ ; we write  $L^*(f, s)$  for the common value.*

*Proof.* Two kinds of variation can change the data: deforming the path shapes with endpoints fixed, and moving the basepoint  $\tau_0$  (which drags the endpoints  $-1/\tau_0, \tau_0 - 1, \tau_0$ ).

*Path shape.* If a segment is deformed within  $\mathbb{H}_f$  with its endpoints fixed and without crossing a pole, the integrand  $f(\tau)K(\tau, s)$  (with  $K \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$ ) is holomorphic on the swept region, so the integral is unchanged by Cauchy's theorem.

*Basepoint.* Differentiating (4.10) in  $\tau_0$  — legitimate by the finiteness of Lemma 4.2 — only the endpoints contribute. The  $T$ -segment endpoints give  $f(\tau_0)\tilde{k}_T(\tau_0, s) - f(\tau_0 - 1)\tilde{k}_T(\tau_0 - 1, s)$ , and the  $S$ -segment endpoints give  $f(\tau_0)\tau_0^{s-1} - f(-1/\tau_0)(-1/\tau_0)^{s-1}\tau_0^{-2}$ . Using the modularity relations  $f(\tau_0 - 1) = f(\tau_0)$  and  $f(-1/\tau_0) = \tau_0^k f(\tau_0)$  together with  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)}\tau_0^{1-s}$ , the total  $\tau_0$ -derivative of  $L^*$  reduces to

$$i^{-s}f(\tau_0)\left[\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) + \tau_0^{s-1} - e^{i\pi(s-1)}\tau_0^{k-1-s}\right] = 0 \quad (4.3)$$

by the shift-difference identity (4.1) of Lemma 4.1. Hence  $L^*$  is constant under any admissible motion of  $\tau_0$ .

Combining the two,  $L^*$  is locally constant on the space of admissible configurations and can change only when a segment crosses a pole: it depends only on the winding data. For  $f \in M_k^!$  there are no interior poles,  $\mathbb{H}_f = \mathbb{H}$  is simply connected, every admissible configuration is joined to every other without crossing a pole, and  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is therefore globally independent of  $\tau_0$  and of the contour pair.  $\square$

### 4.3 The unfolding engine

**Proposition 4.4** (Hurwitz unfolding). *For every nonzero integer  $n$  and every  $\nu \in \mathbb{C}$ ,*

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n), \quad (4.4)$$

where for  $n < 0$  the right side is read on the analytic continuation with  $\arg(2\pi n) = +\pi$ . At the integer values  $\nu \in \{1-s, s-(k-1)\}$  used below (with  $s$  an integer in the critical strip),  $1-\nu$  is a positive integer  $N$  and  $\Gamma(1-\nu, x)$  is the entire polynomial-times-exponential  $\Gamma(N, x) = (N-1)! e^{-x} \sum_{j=0}^{N-1} x^j/j!$ , single-valued for either sign of  $n$ .

*Proof.* For  $\operatorname{Re}(\nu) > 1$  the series  $\zeta(\nu, \sigma) = \sum_{m \geq 0} (\sigma + m)^{-\nu}$  converges uniformly on  $[i, i+1]$  [6, §25.11]; substituting  $u = \sigma + m$ , the periodicity  $e^{2\pi i n m} = 1$  glues the unit segments into a single ray — the *unfolding*, after Rankin–Selberg:

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = \int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du. \quad (4.5)$$

Substituting  $t = -2\pi i n u$  and recalling  $\Gamma(\sigma, x) = \int_x^\infty t^{\sigma-1} e^{-t} dt$  [6, §8.2.2] evaluates the ray integral as

$$\int_i^{i+\infty} e^{2\pi i n u} u^{-\nu} du = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, 2\pi n), \quad (4.6)$$

which is (4.4) for  $\operatorname{Re}(\nu) > 1$  (for  $n < 0$  the path runs to  $+i\infty$ , giving the  $\arg(2\pi n) = +\pi$  continuation of  $\Gamma$ ). Both sides are entire in  $\nu$  ( $\Gamma$  by [6, §8.2.7]; the left integral since the  $\zeta$ -pole at argument 1 integrates to zero over a unit segment), so the identity extends to all  $\nu \in \mathbb{C}$ .  $\square$

The same unfolding at an arbitrary base point  $\tau_0 \in \mathbb{H}$  in place of  $i$  — the translate-gluing  $\sum_{m \geq 0} \int_{\tau_0+m}^{\tau_0+m+1} = \int_{\tau_0}^{\tau_0+\infty}$  is base-point-independent — gives, for every nonzero integer  $n$  and every  $\nu \in \mathbb{C}$ ,

$$\int_{\tau_0}^{\tau_0+1} e^{2\pi i n \sigma} \zeta(\nu, \sigma) d\sigma = i^{1-\nu} (2\pi n)^{\nu-1} \Gamma(1-\nu, -2\pi i n \tau_0), \quad (4.7)$$

whose second argument has  $\operatorname{Re}(-2\pi i n \tau_0) = 2\pi n \operatorname{Im}(\tau_0)$  — positive for  $n > 0$ , negative for  $n < 0$  — with the  $\arg = +\pi$  continuation in the latter case. Equation (4.4) is the case  $\tau_0 = i$ ; we use (4.7) in Section 5.

**Lemma 4.5** (Constant-mode integral). *For every  $s \in \mathbb{C}$ ,*

$$\int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -\frac{i^s}{s} + e^{i\pi(s-1)} \frac{i^{k-s}}{k-s}. \quad (4.8)$$

*Proof.* The Hurwitz zeta has antiderivative  $\int \zeta(\sigma, a) da = -\zeta(\sigma-1, a)/(\sigma-1) + C$  for  $\sigma \neq 1$ ; evaluating across the shifted segment and applying the raising identity  $\zeta(\sigma-1, a+1) - \zeta(\sigma-1, a) = -a^{1-\sigma}$  at  $a = i$  gives

$$\int_{i-1}^i \zeta(\sigma, \tau+1) d\tau = \int_i^{i+1} \zeta(\sigma, u) du = \frac{i^{1-\sigma}}{\sigma-1}. \quad (4.9)$$

Applying (4.9) to the two halves of  $\tilde{k}_T(\tau, s)$  at  $\sigma = 1-s$  and  $\sigma = s-(k-1)$  yields (4.8).  $\square$

#### 4.4 Restated theorem and proof

For reference, the theorem statement is the following.

**Theorem 4.6** (*L*-function for weakly holomorphic forms – Theorem 1.6, restated). *Let  $k \in 2\mathbb{Z}$  and  $f \in M_k^!$ . With  $\tilde{k}_T(\tau, s)$  the Hurwitz-zeta kernel of (1.7), for every  $\tau_0 \in \mathbb{H}$  and every  $s \in \mathbb{C}$ ,*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) := i^{-s} \left[ \int_{\gamma_{\tau_0}^S} f(\tau) \tau^{s-1} d\tau + \int_{\gamma_{\tau_0}^T} f(\tau) \tilde{k}_T(\tau, s) d\tau \right] \quad (4.10)$$

*is finite and independent of both  $\tau_0 \in \mathbb{H}$  and of the admissible contour choices  $\gamma_{\tau_0}^S, \gamma_{\tau_0}^T$ . We write  $L^*(f, s)$  for the common value. Specializing to  $\tau_0 = i$ , the  $S$ -integral in (4.10) vanishes, and  $L^*$  reduces to a single integral on the compact contour  $[i-1, i] \subset \mathbb{H}$ , with closed-form sum*

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_f(n) \left[ \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right]. \quad (4.11)$$

*The right side of (4.11) is the completed  $L$ -function  $L^*(f, s)$  of  $f$ . These  $L$ -integrals reproduce the Hecke-Weil Mellin transform for holomorphic modular forms, as well as reproduce extant formulae from the literature for weakly holomorphic modular forms [2].*

*Proof of Theorem 1.6, aka Theorem 4.6.* The proof runs in seven steps: well-definedness (Steps 1–2), the collapse at  $\tau_0 = i$  (Step 3), the per-mode evaluation (Steps 4–6), and the identification (Step 7).

**Step 1: finiteness.** For  $f \in M_k^!$  one has  $\mathbb{H}_f = \mathbb{H}$ , so every  $\tau_0 \in \mathbb{H}$  gives an admissible contour pair and both integrals in (4.10) are finite for every  $s \in \mathbb{C}$  by Lemma 4.2.

**Step 2:  $\tau_0$ - and contour-independence.** By Lemma 4.3, specialised to the simply-connected  $\mathbb{H}_f = \mathbb{H}$  of a weak form,  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\tau_0$  and of the admissible contour pair. This establishes the well-definedness assertion of (4.10) and licenses writing  $L^*(f, s)$  for the common value.

**Step 3: collapse at  $\tau_0 = i$ .** At the elliptic fixed point  $\tau_0 = i$  the  $S$ -segment endpoints coincide,  $-1/i = i$ , so  $\gamma_i^S$  has zero length and the  $S$ -integral in (4.10) vanishes. By Step 2 the value is unchanged, leaving

$$L^*(f, s) = i^{-s} \int_{i-1}^i f(\tau) \tilde{k}_T(\tau, s) d\tau. \quad (4.12)$$

**Step 4: per-mode expansion.** Expanding  $f = \sum_n c_f(n) q^n$  in (4.12) and swapping sum and integral — justified by uniform convergence of the  $q$ -series on the compact segment  $[i-1, i]$ , including the finitely many polar modes, since  $|q^{-1}| = e^{2\pi}$  is bounded there — gives

$$L^*(f, s) = i^{-s} \sum_n c_f(n) \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau. \quad (4.13)$$

**Step 5: the constant mode.** For  $n = 0$ , Lemma 4.5 and the phase collapse  $i^{-s} e^{i\pi(s-1)} i^{k-s} = -i^k$  give the contribution

$$c_f(0) \cdot i^{-s} \int_{i-1}^i \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right), \quad (4.14)$$

the polar term of [3, Lemma 2.1] (equivalently the polar piece of [2, Thm. 1.1]) at the splitting point  $B = 1$ .

**Step 6: the non-constant modes.** For  $n \neq 0$ , substituting  $\sigma = \tau + 1$  shifts the contour to  $[i, i+1]$  with prefactor  $e^{-2\pi i n} = 1$ , and Proposition 4.4 applied to the two halves of  $\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1)$  at  $\nu = 1-s$  and  $\nu = s-(k-1)$  gives

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(1-s, \sigma) d\sigma = i^s (2\pi n)^{-s} \Gamma(s, 2\pi n), \quad (4.15)$$

$$\int_i^{i+1} e^{2\pi i n \sigma} \zeta(s-(k-1), \sigma) d\sigma = i^{k-s} (2\pi n)^{s-k} \Gamma(k-s, 2\pi n). \quad (4.16)$$

The first half against the  $i^{-s}$  prefactor gives  $\Gamma(s, 2\pi n)/(2\pi n)^s$ ; the second against the relative phase  $-e^{i\pi(s-1)}$  collapses via  $-i^{-s} e^{i\pi(s-1)} i^{k-s} = i^k$  to  $i^k \Gamma(k-s, 2\pi n)/(2\pi n)^{k-s}$ . Hence the per-mode summand is

$$c_f(n) \cdot i^{-s} \int_{i-1}^i e^{2\pi i n \tau} \tilde{k}_T(\tau, s) d\tau = c_f(n) \left[ \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right], \quad (4.17)$$

the  $B = 1$  value of the cuspidal summand of [3, Lemma 2.1]. Summing (4.14) and (4.17) over all Fourier modes in (4.13) reproduces the closed-form sum (4.11).

**Step 7: identification.** The right side of (4.11) is the completed  $L$ -function  $L^*(f, s)$ : for holomorphic  $f$  it is the classical Hecke–Weil Mellin transform, and for weak  $f$  it is the incomplete-gamma sum of [2], here obtained at every  $s \in \mathbb{C}$  from a finite contour with no regulator. This completes the proof.  $\square$

## 4.5 Worked example at $k = 12$ : Bernoulli polynomials and numerics

At  $k = 12$  the kernel of (1.7) is

$$\tilde{k}_T(\tau, s) = \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-11, \tau+1). \quad (4.18)$$

At integer values of  $s \in \{1, 2, \dots, 11\}$ , both Hurwitz zeta factors collapse to Bernoulli polynomials via  $\zeta(-m, a) = -B_{m+1}(a)/(m+1)$  ( $m \geq 0$ ), and  $\tilde{k}_T(\tau, s)$  becomes a polynomial of degree  $\leq 11$  in  $\tau$ . Three representative values:

$$\tilde{k}_T(\tau, 1) = -\tau - \frac{1}{2} + \frac{1}{11} B_{11}(\tau+1), \quad \tilde{k}_T(\tau, 6) = -\frac{1}{3} B_6(\tau+1), \quad (4.19)$$

$$\tilde{k}_T(\tau, 11) = -\frac{1}{11} B_{11}(\tau+1) + \tau + \frac{1}{2}. \quad (4.20)$$

The  $s \leftrightarrow 12-s$  symmetry follows from the  $s \leftrightarrow k-s$  structure of the two-term kernel (with  $i^k = +1$  at  $k = 12$ ).

**Identification with Theorem 1.4.** With the clean form of Theorem 1.4, the  $T$ -segment kernel of (1.9) is exactly the integer- $s = \ell$  specialisation  $\tilde{k}_T(\tau, \ell)$  of the Hurwitz-zeta kernel of (1.7), with no convention conversion required: all combinatorial structure  $(-1)^{\ell-1} \binom{k-2}{\ell-1} / c_S^{\ell-1}$  governing the  $\omega^\pm(f)$  normalisation lives outside the two-segment bracket. The period-extraction functional (1.9) is thus the rescaled integer- $s$  shadow of the all- $s$   $L$ -function functional (4.10) of Theorem 1.6, with the bare Mellin  $\tau^{s-1}$  and Hurwitz  $\tilde{k}_T(\tau, s)$  kernels mapped to their integer- $s = \ell$  specialisations and the analytic prefactor  $i^{-s}$  replaced by the combinatorial Brown-basis factor.

**Numerical values.** The companion `mpmath` scripts [4], run at 50-digit precision with  $f \in \{\Delta, \hat{\Delta}\}$ , reproduce the periods  $\omega^\pm(\Delta)$  and quasi-periods  $\eta^\pm(\hat{\Delta})$  of [1] listed in Table 1, to relative residual  $\lesssim 10^{-40}$ , the same for every interior  $\ell \in \{2, \dots, 10\}$  and every basepoint  $\tau_0 \in \{0.3+1.2i, i, 1.5i, -0.4+0.8i\}$ .

**Complex- $s$  verification.** Beyond the integer- $\ell$  period extraction, the all- $s$  functional (4.10) was checked directly against the canonical completed  $L$ -function  $L^*(\Delta, s)$  of Theorem 1.6 at non-integer  $s$  [4]. At half-integer points  $s \in \{1.5, 2.5, \dots, 6.5\}$ , off-axis complex points  $s \in \{2.7 + 0.4i, 5 + i, 5.5 + 2.5i\}$ , and points outside the critical strip  $s \in \{0.5, -0.3\}$ , the finite-contour value agrees with  $L^*(\Delta, s)$  to relative residual  $\sim 10^{-53}$  at 50-digit precision, and is independent of  $\tau_0 \in \{0.3 + 1.2i, 1.5i, -0.4+0.8i, 0.2+2i\}$  at the same level. The corresponding check for  $\hat{\Delta}$  holds to  $\sim 10^{-47}$ , the loss of digits reflecting the  $q^{-1}$  cusp mode.

#### 4.6 Classical Mellin recovery and canonical uniqueness.

For  $f \in S_k$  a holomorphic cusp form, taking the cusp-limit  $\tau_0 \rightarrow i\infty$  in (4.10) collapses the  $S$ -segment onto the classical Mellin contour  $[0, i\infty]$  while the  $T$ -segment vanishes exponentially (since  $f(\tau) = O(e^{-2\pi \text{Im } \tau})$  on the segment as  $\text{Im } \tau \rightarrow \infty$ ):

$$\lim_{\tau_0 \rightarrow i\infty} L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \int_0^\infty t^{s-1} f(it) dt, \quad (4.21)$$

the classical completed  $L$ -integral of  $f$ . For  $f \in M_k$  with  $c_f(0) \neq 0$  the same limit holds after a standard constant-term subtraction (see [3, Lemma 2.1] for the explicit polynomial correction).

For  $f \in M_k^!$  with cusp poles the cusp-limit is genuinely pathological:  $f(\tau) \rightarrow \infty$  as  $\text{Im } \tau \rightarrow \infty$ , so the  $S$ -segment integral diverges along the imaginary axis, and the  $T$ -segment integral  $\int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau$  acquires a matching divergence of the opposite sign as  $\text{Im } \tau_0 \rightarrow \infty$  — made precise in Lemma 5.4, where the  $T$ -segment is shown to equal an explicit incomplete-gamma sum whose polar mode  $q^{-M}$  contributes a term growing like  $|c_f(-M)| e^{2\pi M \text{Im } (\tau_0)}$ . The two divergences cancel at every finite  $\tau_0$ , so  $L^*(f, s)$  is finite and well defined throughout  $\mathbb{H}$ , but no naive Mellin contour limit exists. By throwing-out  $\gamma_{\tau_0}^T$  at the outset, one does violence to the result; the  $\tau_0 \rightarrow i\infty$  limit of the  $S$ -contour alone diverges, necessitating regularizations of the type introduced in [2]. The construction of [2] recovers  $L^*(f, s)$  on  $M_k^!$  by a non-holomorphic regulator at the cusps; Theorem 1.6 shows that this regulator and the finite-contour functional (4.10) compute the same object, term by term in the Fourier expansion. In particular,  $L^*(f, s)$  is independent of any choice of regularisation: it is the value of the finite-contour functional at any admissible  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ , and the BFK construction is one closed-form evaluation of it (at  $\tau_0 = i$ ).

## 5 Wall-crossing for meromorphic forms: proof of Theorem 1.7

### 5.1 Overview

Theorem 1.7 has three parts. Parts (a) (finiteness) and (b) (dependence on the admissible contour pair only through its homotopy class in  $\mathbb{H}_f$ ) hold for arbitrary meromorphic  $f$  and were already proved, without any cuspidality hypothesis, as Lemma 4.2 and Lemma 4.3 of Section 4 — stated there in exactly this generality so as to serve here. The new content of this section is part (c): the explicit residue  $\mathcal{R}_{\tau_p}(f, s)$  by which  $L^*$  jumps when the contour pair crosses an interior pole (Lemma 5.1, Proposition 5.2), assembled into the proof of Theorem 1.7 below. We then reduce  $L^*$  to the deformed-Mellin  $L$ -function  $L_f^*(s)_{\text{int}}$  of [3] on forms regular at the cusp (Corollary ??), record the integer- $s$  period-polynomial form of the wall-crossing (Corollary 5.9), and state the per-mode closed form for interior-pole forms regular at the cusp (Proposition 5.10).

### 5.2 The wall-crossing residue

We compute the jump in  $L^*$  when a contour deformation drags one segment across a single interior pole.

**Lemma 5.1** (Single-segment crossing). *Let  $f$  be a meromorphic modular form of weight  $k$  with a simple pole at  $\tau_p \in \mathbb{H}$ , and let  $K(\tau, s) \in \{\tau^{s-1}, \tilde{k}_T(\tau, s)\}$  be either kernel. Suppose a continuous deformation carries a segment from  $\Sigma_0$  to  $\Sigma_1$  (within  $\mathbb{H}_f$  away from  $\tau_p$ ) so that the closed loop  $\Sigma_1 - \Sigma_0$  encircles  $\tau_p$  once with orientation  $\sigma \in \{+1, -1\}$  and no other pole. Then*

$$\int_{\Sigma_1} f(\tau) K(\tau, s) d\tau - \int_{\Sigma_0} f(\tau) K(\tau, s) d\tau = 2\pi i \sigma \text{Res}(f, \tau_p) K(\tau_p, s). \quad (5.1)$$

*Proof.* The kernel  $K(\cdot, s)$  is analytic on  $\mathbb{H}$ , so  $f(\tau) K(\tau, s)$  has a simple pole at  $\tau_p$  with residue  $\text{Res}(f, \tau_p) K(\tau_p, s)$  and no other singularity in the region swept between  $\Sigma_0$  and  $\Sigma_1$ . The difference of the two integrals is the integral over the closed loop  $\Sigma_1 - \Sigma_0$ , which by the residue theorem equals  $2\pi i \sigma \text{Res}(f, \tau_p) K(\tau_p, s)$ .  $\square$

**Proposition 5.2** (Wall-crossing residue). *Let  $f$  be a meromorphic modular form of weight  $k$ , and let  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and  $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$  be two admissible contour pairs whose homotopy classes in  $\mathbb{H}_f$  differ by a single crossing of a simple interior pole  $\tau_p \in \mathbb{H}$  of  $f$ , with  $S$ - and  $T$ -segment winding numbers  $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$ . Then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \left[ \sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (5.2)$$

which is the residue  $\mathcal{R}_{\tau_p}(f, s)$  of (1.13).

*Proof.* Apply Lemma 5.1 to the  $S$ -segment with kernel  $\tau^{s-1}$  and winding  $\sigma_{\tau_p}^S$ , and to the  $T$ -segment with kernel  $\tilde{k}_T(\tau, s)$  and winding  $\sigma_{\tau_p}^T$ . Carried through the common prefactor  $i^{-s}$  of (1.10), the two contributions sum to (5.2). At a generic wall exactly one of  $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T$  is nonzero, according to which segment crosses; both are nonzero only on the codimension-two locus where the two segments meet  $\Gamma$ -images of  $\tau_p$  simultaneously.  $\square$

For a pole of order  $m \geq 1$  the same computation applies verbatim, with  $\text{Res}(f, \tau_p) K(\tau_p, s)$  in (5.1) replaced by the order- $m$  residue  $\sum_{\ell=1}^m r_{f, \tau_p}(-\ell) \partial_{\tau_p}^{\ell-1} K(\tau_p, s) / (\ell-1)!$ , where  $r_{f, \tau_p}(-\ell)$  are the principal-part Laurent coefficients of  $f$  at  $\tau_p$ ; the generalization is routine, and we record only the simple-pole case.

### 5.3 Restated theorem and proof

For reference, the theorem statement is the following.

**Theorem 5.3** (*L-function for meromorphic modular forms – Theorem 1.7, restated*). *Let  $k \in 2\mathbb{Z}$  and let  $f$  be a meromorphic modular form of weight  $k$  on  $\Gamma$ . Write  $\mathbb{H}_f := \mathbb{H} \setminus \Gamma \cdot f^{-1}(\infty)$  for the upper half-plane with the  $\Gamma$ -orbit of the interior poles of  $f$  removed, and call the pair  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  admissible if  $\tau_0 \in \mathbb{H}_f$  and both contours lie in  $\mathbb{H}_f$ , with  $\gamma_{\tau_0}^S$  running from  $S^{-1}\tau_0 = -1/\tau_0$  to  $\tau_0$  and  $\gamma_{\tau_0}^T$  from  $T^{-1}\tau_0 = \tau_0 - 1$  to  $\tau_0$ . Define  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  by (1.10) with the Hurwitz-zeta kernel  $\tilde{k}_T(\tau, s)$  of (1.7). Then:*

- (a) *Both integrals in (1.10) are finite for every admissible  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and every  $s \in \mathbb{C}$ .*
- (b)  *$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  depends on the admissible pair only through its homotopy class in  $\mathbb{H}_f$ : any continuous one-parameter family of admissible pairs — including variations of the anchoring basepoint  $\tau_0$  — leaves the value unchanged.*
- (c) *If two admissible configurations  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  and  $(\gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T)$  lie in homotopy classes related by a single crossing of an interior pole  $\tau_p \in \mathbb{H}$  of  $f$ , then*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) - L^*(f, s; \gamma_{\tau'_0}^S, \gamma_{\tau'_0}^T) = \mathcal{R}_{\tau_p}(f, s), \quad (5.3)$$

where for a simple pole  $\text{Res}(f, \tau_p)$ ,

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \text{Res}(f, \tau_p) \left[ \sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s) \right], \quad (5.4)$$

with  $\sigma_{\tau_p}^S, \sigma_{\tau_p}^T \in \{-1, 0, +1\}$  the winding-number contributions of  $\gamma_{\tau_0}^S$  and  $\gamma_{\tau_0}^T$  across  $\tau_p$  (the generic case has one nonzero and the other zero, according to which contour crossed). Higher-order poles contribute via the corresponding derivatives in  $\tau_p$  of (5.4), as recorded after Proposition 5.2.

*Proof of Theorem 1.7, aka Theorem 5.3.* (a) Finiteness of both integrals for any admissible pair is Lemma 4.2. (b) Dependence on the admissible pair only through its homotopy class in  $\mathbb{H}_f$  — invariance under any admissible deformation, the basepoint included — is Lemma 4.3. (c) When two admissible classes differ by a single crossing of a simple interior pole  $\tau_p$ , the jump (5.3) is (5.2) of Proposition 5.2, equal to  $\mathcal{R}_{\tau_p}(f, s)$  of (5.4); the order- $m$  case is the higher residue recorded after Proposition 5.2.  $\square$

### 5.4 Reduction on regular-at-cusps meromorphic forms

The two-segment functional makes the cusp behaviour of the  $T$ -segment completely explicit: it is the BFK incomplete-gamma sum with the split height promoted to the endpoint  $\tau_0$ . We record this closed form and its asymptotics, then specialise to forms regular at the cusp to reduce  $L^*$  to the deformed-Mellin  $L$ -function  $L_f^*(s)_{\text{int}}$  of [3].

**Lemma 5.4** (*T-segment closed form and cusp asymptotics*). *Let  $f \in M_k^!$  with Fourier expansion  $f = \sum_n c_f(n) q^n$ . For every  $\tau_0 \in \mathbb{H}$  and every  $s \in \mathbb{C}$ ,*

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = -c_f(0) \left[ \frac{(\tau_0/i)^s}{s} + i^k \frac{(\tau_0/i)^{k-s}}{k-s} \right] + \sum_{n \neq 0} c_f(n) \left[ \frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (5.5)$$

Consequently, as  $\text{Im}(\tau_0) \rightarrow \infty$  with  $\text{Re}(\tau_0)$  fixed:

- (a) if  $f$  is regular at the cusp with  $f = O(q^N)$ ,  $N \geq 1$ , the  $T$ -segment integral decays as  $O(e^{-2\pi N \operatorname{Im}(\tau_0)})$  (up to a polynomial in  $\tau_0$ );
- (b) if  $f$  has a cusp pole  $f = c_f(-M)q^{-M} + \dots$  with  $c_f(-M) \neq 0$ ,  $M \geq 1$ , the  $T$ -segment integral grows as  $|c_f(-M)|e^{2\pi M \operatorname{Im}(\tau_0)}$ , matching the cusp-pole growth of  $f$  itself.

*Proof.* Expand  $f = \sum_n c_f(n)q^n$  on the compact segment  $[\tau_0 - 1, \tau_0]$  (uniformly convergent — the finitely many polar modes are bounded there,  $|q|$  being bounded below) and integrate term by term. For  $n \neq 0$ , the base- $\tau_0$  unfolding (4.7) applied to the two halves of  $\tilde{k}_T(\tau, s) = \zeta(1 - s, \tau + 1) - e^{i\pi(s-1)}\zeta(s - (k-1), \tau + 1)$  at  $\nu = 1 - s$  and  $\nu = s - (k-1)$ , with the phase collapse  $-i^{-s}e^{i\pi(s-1)}i^{k-s} = i^k$  of Section 4.4, gives the  $n \neq 0$  summand of (5.5). For  $n = 0$ , the Hurwitz antiderivative of Lemma 4.5 at base  $\tau_0$ ,  $\int_{\tau_0-1}^{\tau_0} \zeta(\sigma, \tau + 1) d\tau = \tau_0^{1-\sigma}/(\sigma - 1)$ , produces the bracketed constant-mode term (which at  $\tau_0 = i$  reduces to the  $-c_f(0)(1/s + i^k/(k-s))$  of (1.11)).

For the asymptotics, the constant-mode term is a polynomial in  $\tau_0$ , hence sub-exponential. In each  $n \neq 0$  summand the incomplete-gamma argument satisfies  $\operatorname{Re}(-2\pi i n \tau_0) = 2\pi n \operatorname{Im}(\tau_0)$ , and the uniform asymptotic  $\Gamma(\sigma, z) \sim z^{\sigma-1}e^{-z}$  as  $|z| \rightarrow \infty$  for  $|\arg z| < \frac{3\pi}{2}$  [6, §8.11] gives  $|c_f(n)|$ -weighted decay  $e^{-2\pi n \operatorname{Im}(\tau_0)}$  for  $n > 0$  and growth  $e^{2\pi |n| \operatorname{Im}(\tau_0)}$  for  $n < 0$ . Taking the lowest present mode yields (a) and (b).  $\square$

For  $f \in F_k$  regular at the cusp ( $f = \sum_{n \geq N} c_f(n)q^n$ ,  $N \geq 0$ ), part (a) of Lemma 5.4 drives the  $T$ -segment to zero as  $\operatorname{Im} \tau_0 \rightarrow \infty$ , which gives the reduction to  $L_f^*(s)_{\text{int}}$ .

**Corollary 5.5** ( $L^*$  on regular-at-cusps meromorphic forms – Corollary ??, restated). *Let  $f \in F_k$  be a meromorphic modular form of weight  $k$  that is regular at the cusp. Let  $L_f^*(s)_{\text{int}, \gamma}$  denote the deformed-Mellin  $L$ -function of  $f$  along a contour  $\gamma$  from 0 to  $i\infty$  lying in  $\mathbb{H}_f$  (see [3, §2]). Then for every admissible  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  whose homotopy class in  $\mathbb{H}_f$  is compatible with  $\gamma$  (in the sense that the contour  $\gamma_{\tau_0}^S$  together with the cusp-limit piece  $\lim_{\tau_0 \rightarrow i\infty} \gamma_{\tau_0}^S \rightarrow [0, i\infty]$  can be deformed to  $\gamma$  in  $\mathbb{H}_f$ ),*

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = L_f^*(s)_{\text{int}, \gamma} \quad \text{for every } s \in \mathbb{C}. \quad (5.6)$$

*Proof of Corollary ??, aka Corollary 5.5.* Take  $\tau_0 = \varepsilon + i\Lambda$  with  $\varepsilon \in \mathbb{R}$  chosen so that  $\operatorname{Re}(\tau) = \varepsilon$  avoids every  $\Gamma$ -image of an interior pole of  $f$ , fixing the homotopy class compatible with  $\gamma$ . By Lemma 4.3 the value  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  is independent of  $\Lambda$  within that class. As  $\Lambda \rightarrow \infty$ , the  $T$ -segment integral tends to 0 by Lemma 5.4 (for  $N \geq 1$ ; for  $N = 0$  the same holds after the constant-term subtraction of [3, Lemma 2.1]), while the  $S$ -segment  $[-1/\tau_0, \tau_0]$  approaches the deformed-Mellin contour  $\gamma_\varepsilon$  along  $\operatorname{Re}(\tau) = \varepsilon$  from 0 to  $i\infty$ . Hence

$$L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = \lim_{\Lambda \rightarrow \infty} L^*(f, s; \gamma_{\varepsilon+i\Lambda}^S, \gamma_{\varepsilon+i\Lambda}^T) = L_f^*(s)_{\text{int}, \gamma_\varepsilon}, \quad (5.7)$$

which is (5.6).  $\square$

## 5.5 Special values and the bare-Mellin residue

The wall-crossing residue  $\mathcal{R}_{\tau_p}(f, s)$  of (5.4) inherits its  $s$ -dependence from the two kernels; only the  $T$ -kernel is singular in  $s$ .

**Lemma 5.6** ( $s$ -poles of the  $T$ -kernel). *For fixed  $\tau_p \in \mathbb{H}$  and  $k \in 2\mathbb{Z} \setminus \{0\}$ , the kernel  $\tilde{k}_T(\tau_p, s)$  of (1.7) is meromorphic in  $s$  with exactly two simple poles, at  $s = 0$  and  $s = k$ , with*

$$\operatorname{Res}_{s=0} \tilde{k}_T(\tau_p, s) = -1, \quad \operatorname{Res}_{s=k} \tilde{k}_T(\tau_p, s) = +1, \quad (5.8)$$

*both independent of  $\tau_p$ ; the bare-Mellin kernel  $\tau_p^{s-1}$  is entire in  $s$ .*

*Proof.* The Hurwitz zeta  $\zeta(\nu, a)$  has a single simple pole, at  $\nu = 1$ , of residue 1 independent of  $a$  [6, §25.11]. In  $\tilde{k}_T(\tau_p, s) = \zeta(1-s, \tau_p+1) - e^{i\pi(s-1)}\zeta(s-(k-1), \tau_p+1)$ , the first term is singular only at  $1-s=1$ , i.e.  $s=0$ , where  $\zeta(1-s, \cdot) = -1/s + O(1)$  gives residue  $-1$ ; the second only at  $s-(k-1)=1$ , i.e.  $s=k$ , where  $\zeta(s-k+1, \cdot) = 1/(s-k) + O(1)$  and the prefactor  $-e^{i\pi(k-1)} = +1$  ( $k$  even) give residue  $+1$ . Both residues are  $\tau_p$ -independent, and  $\tau_p^{s-1} = e^{(s-1)\log \tau_p}$  is entire.  $\square$

**Corollary 5.7** (*s*-poles of the wall-crossing residue). *The residue  $\mathcal{R}_{\tau_p}(f, s)$  of (5.4) is entire in  $s$  in its  $S$ -segment part, while its  $T$ -segment part has simple poles only at  $s=0, k$ ,*

$$\text{Res}_{s=0} \mathcal{R}_{\tau_p} = +2\pi i \sigma_{\tau_p}^T \text{Res}(f, \tau_p), \quad \text{Res}_{s=k} \mathcal{R}_{\tau_p} = -2\pi i i^{-k} \sigma_{\tau_p}^T \text{Res}(f, \tau_p). \quad (5.9)$$

*In particular  $\mathcal{R}_{\tau_p}$  is entire in  $s$  if and only if  $\sigma_{\tau_p}^T = 0$ .*

*Proof.* Immediate from Lemma 5.6 and the form  $\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \text{Res}(f, \tau_p) [\sigma_{\tau_p}^S \tau_p^{s-1} + \sigma_{\tau_p}^T \tilde{k}_T(\tau_p, s)]$ , with  $i^{-s} = 1$  at  $s=0$ .  $\square$

**Remark.** The constant Fourier mode contributes the special-value poles  $-c_f(0)(1/s + i^k/(k-s))$  of (1.11). By Corollary 5.7 an  $S$ -segment crossing shifts  $L^*$  by an amount entire in  $s$  and so cannot disturb these special values, whereas a  $T$ -segment crossing can. This is the finite-contour reading of the statement of [3, §2.4] that the special values are insensitive to the wall-crossing ambiguity: it is exactly the assertion  $\sigma_{\tau_p}^T = 0$  at every crossed pole, which holds in the cusp-limit class of Proposition 5.8.

**Proposition 5.8** (Bare-Mellin reduction in the cusp limit). *Let  $f \in F_k$  be regular at the cusp, and let two admissible contour pairs in the cusp-limit class of Corollary 5.5 differ by a single crossing of an interior pole  $\tau_p$ . Then  $\sigma_{\tau_p}^T = 0$  — the  $T$ -segment has retreated to the cusp, where its  $\tilde{k}_T$ -weighted integral vanishes by Lemma 5.4(a) — so the jump is the single bare-Mellin term*

$$\mathcal{R}_{\tau_p}(f, s) = -2\pi i i^{-s} \sigma_{\tau_p}^S \text{Res}(f, \tau_p) \tau_p^{s-1}, \quad (5.10)$$

*entire in  $s$ . This is the single-pole term of the deformed-Mellin wall-crossing [3, Eq. (2.65)] under the dictionary  $\chi(\tau_p) = -\sigma_{\tau_p}^S$ , with the bare-Mellin residue  $i^{-s} \text{Res}(f, \tau_p) \tau_p^{s-1}$  of the integrand  $i^{-s} f(\tau) \tau^{s-1}$ .*

*Proof.* In the cusp-limit class the  $T$ -segment is sent to the cusp; for  $f$  regular there its  $\tilde{k}_T$ -integral vanishes by Lemma 5.4(a), so no  $T$ -crossing occurs and  $\sigma_{\tau_p}^T = 0$ . Setting  $\sigma_{\tau_p}^T = 0$  in (5.4) gives (5.10), entire in  $s$  by Corollary 5.7; the match with [3, Eq. (2.65)] is Corollary 5.5 applied to the residue.  $\square$

*Honest weight.* Proposition 5.8 is a consistency repackaging of Corollary 5.5 and the residue theorem; the genuinely new content is Lemma 5.6 and its special-value consequence, Corollary 5.7.

## 5.6 Wall-crossing of the period-polynomial cocycle

The wall-crossing residue formula (1.13) acts on the complex- $s$  functional  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$ . Specialised at integer  $s = \ell \in \{2, \dots, k-2\}$  and assembled into the  $V_{k-2}$  moment expansion of Section 2, it gives the wall-crossing of the period-polynomial cocycle of  $f$  under contour deformation across an interior pole.

**Corollary 5.9** (Period-polynomial wall-crossing residue). *Let  $f$  be a meromorphic modular form of weight  $k$  on  $\Gamma$ , and let  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T), (\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  be two admissible contour pairs related by a single crossing of an interior pole  $\tau_p \in \mathbb{H}$  of  $f$ . Denote by  $C_S^f(\gamma_{\tau_0}^S), C_T^f(\gamma_{\tau_0}^T) \in V_{k-2} \otimes \mathbb{C}$  the corresponding  $V_{k-2}$ -valued cocycle values from (2.3). For a simple pole  $\text{Res}(f, \tau_p)$ ,*

$$C_S^f(\gamma_{\tau_0}^S) - C_T^f(\gamma_{\tau_0}^T) = 2\pi i (2\pi i)^{k-2} \sigma_{\tau_p}^S \text{Res}(f, \tau_p) (X - \tau_p Y)^{k-2}, \quad (5.11)$$

and analogously for  $C_T^f$  with  $\sigma_{\tau_p}^S$  replaced by  $\sigma_{\tau_p}^T$ . The integer- $s = \ell$  specialisation of (1.13) is the coefficient of  $\binom{k-2}{\ell} (-1)^\ell X^{k-2-\ell} Y^\ell$  in (5.11) (combined for  $S$  and  $T$ ). For  $\dim S_k = 1$  and  $f \in F_k$  meromorphic with interior poles, the period polynomial  $r_f$  extracted via Theorem 1.4 from  $(\gamma_{\tau_0}^S, \gamma_{\tau_0}^T)$  therefore depends on the homotopy class of the contour pair, with jump (5.11) across each wall.

*Proof.* The integrand  $f(\tau)(X - \tau Y)^{k-2}$  of (2.3) has a simple pole at  $\tau = \tau_p$  with residue  $\text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$ , so a contour deformation crossing  $\tau_p$  with winding  $\sigma_{\tau_p} \in \{-1, +1\}$  contributes  $2\pi i \sigma_{\tau_p} \text{Res}(f, \tau_p)(X - \tau_p Y)^{k-2}$  to the moment integral. Multiplying by the cocycle prefactor  $(2\pi i)^{k-2}$  of (2.3) gives (5.11). Expanding  $(X - \tau_p Y)^{k-2}$  in the monomial basis and matching coefficients of  $X^{k-2-\ell} Y^\ell$  for each  $\ell \in \{0, \dots, k-2\}$  reproduces (1.13) at integer  $s = \ell$  (with  $\tau_p^{\ell-1}$  arising from  $X^{k-2-\ell} Y^\ell$  at  $\ell \geq 1$  from the  $S$ -segment contribution, and analogously for the  $T$ -segment via  $\tilde{k}_T(\tau_p, \ell)$  at integer  $\ell$ ).  $\square$

## 5.7 Remark on the polylog-projection machinery

The polylog projection  $\hat{P}_{\tau_p} : f \mapsto \sum_{m>0} r_{f, \tau_p}^*(m) \text{Li}_{1-m}(e(\tau - \tau_p))$  (see [3, §2.2]) subtracts the leading polar tails of  $f$  at each interior pole  $\tau_p$ , ensuring that the  $\Lambda$ -truncated Mellin integral against  $f$  has a finite  $\Lambda \rightarrow \infty$  limit (necessary in the deformed-Mellin construction because of the Hagedorn-rate Fourier coefficients  $|c_f(n)| \sim n^{p-1} e^{2\pi n \text{Im}(\tau_p)}$  generated by interior poles [3, Prop. 1.1, 2.7]). In the finite-contour formulation (1.10) the  $\Lambda \rightarrow \infty$  limit is not taken, so  $\hat{P}_{\tau_p}$  is not required for the existence or local-constancy properties of  $L^*$ . The polylog-projection machinery does remain relevant, however, for the *closed-form per-mode evaluation* of  $L^*$  in incomplete gammas parallel to (1.11), where the Hagedorn-rate Fourier modes must be split off and dealt with separately before each mode contributes a clean incomplete-gamma term. For forms regular at the cusp this per-mode evaluation is available in closed form, as follows.

**Proposition 5.10** (Per-mode closed form for interior-pole forms regular at the cusp). *Let  $f \in F_k$  be regular at the cusp, with interior poles at the points  $\tau_p$  of a fundamental domain and principal-part data  $r_{f, \tau_p}^*(m)$ . Then, for every admissible contour pair in the homotopy class of Corollary ??,  $L^*(f, s)$  has the per-mode closed form*

$$\begin{aligned} L^*(f, s) = & -c_f(0) \left( \frac{1}{s} + \frac{i^k}{k-s} \right) + \sum_{n \neq 0} c_{Rf}(n) \left[ \frac{\Gamma(s, 2\pi n)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, 2\pi n)}{(2\pi n)^{k-s}} \right] \\ & + \sum_{\tau_p} \sum_{m>0} r_{f, \tau_p}^*(m) [I_{m-1}(s, \tau_p) + i^k I_{m-1}(k-s, \tau_p)], \end{aligned} \quad (5.12)$$

where  $c_{Rf}(n)$  are the Fourier coefficients of the polylog-regularised part  $(1 - \hat{P})f$  (with  $\hat{P} = \sum_{\tau_p} \hat{P}_{\tau_p}$ ), and  $I_N(s, \tau_p)$  are the polylog integrals of [3, §2].

We state Proposition 5.10 without reproducing its derivation: it is the  $B = 1$  specialisation of the  $L$ -function of [3, §2] for  $L_f^*(s)_{\text{int}}$ , obtained there by the polylog projection  $\hat{P}_{\tau_p}$ , and is inherited here through Corollary ??. The genuinely new regime —  $f$  with poles *both* in  $\mathbb{H}$  and at the cusp — is reached by combining the interior projection with the finite-contour cusp treatment of Section 4, through a Hauptmodul split that keeps both pieces modular.

**Lemma 5.11** (Hauptmodul split). *Let  $k \in 2\mathbb{Z}$  with  $S_{2-k} = \{0\}$  — equivalently  $k \geq 0$ , or  $k \in \{-2, -4, -6, -8, -12\}$  — and let  $f \in F_k$ . Then  $f$  splits as  $f = g + h$ , with  $g \in M_k^!$  having poles only at the cusp and  $h$  regular at the cusp, carrying exactly  $f$ 's interior principal parts. For  $k \geq 4$  the split is explicit:  $f/E_k$  is a weight-0 modular function ( $E_k = 1 + O(q)$  at the cusp), hence rational in the Hauptmodul  $j$ , and*

$$f = \underbrace{E_k P(j)}_g + \underbrace{\sum_{\tau_p} \sum_{l \geq 1} a_{\tau_p, l} \frac{E_k}{(j - j(\tau_p))^l}}_h \quad (5.13)$$

is its polynomial part plus the partial fractions over the interior poles  $\tau_p$ , each  $E_k/(j - j(\tau_p))^l$  being regular at the cusp ( $O(q^l)$  there). Equivalently, clearing the interior poles by  $\Phi = \prod_{\tau_p} (j - j(\tau_p))^{m_p}$  gives  $f\Phi \in M_k^!$ , and (5.13) is the  $j$ -partial-fraction of  $f = (f\Phi)/\Phi$ . Both  $g$  and  $h$  are weight- $k$  modular.

Role of  $S_{2-k}$ . What the split needs is that  $f$ 's cusp principal part be realizable in  $M_k^!$ , and this is exactly the condition  $S_{2-k} = \{0\}$ . Indeed, for  $G \in S_{2-k}$  the product  $f G d\tau$  has weight  $k + (2 - k) = 2$  and so is a meromorphic differential on  $X(1)$ , whose residues sum to zero:

$$\text{Res}_\infty(f G d\tau) = - \sum_{\tau_p} \text{Res}_{\tau_p}(f G d\tau). \quad (5.14)$$

The cusp principal part is realizable — equivalently the split exists — iff the left-hand side vanishes for every  $G \in S_{2-k}$ , which is automatic when  $S_{2-k} = \{0\}$ . At the exotic weights  $k = -10$  and  $k \leq -14$ , where  $S_{2-k} \neq \{0\}$ , (5.14) is a genuine constraint that can force  $h$  to retain a cusp pole; there one evaluates  $L^*(f)$  directly from the finite contour (still finite by Lemma 4.2). (At  $\tau_p \in \{i, \rho\}$  the powers  $l$  follow the ramification of  $j$ .)

**Proposition 5.12** (Per-mode closed form, poles interior and at the cusp). *Let  $k \in 2\mathbb{Z}$  with  $S_{2-k} = \{0\}$  and  $f \in F_k$  have poles both in  $\mathbb{H}$  and at the cusp. Then  $L^*(f, s)$  is finite for every  $s \in \mathbb{C}$  at every admissible  $\tau_0$  (Lemma 4.2) and has the per-mode closed form (5.12), where now  $c_{Rf}(n)$  are the Fourier coefficients of the cusp part  $g \in M_k^!$  of Lemma 5.11, including its cusp-polar modes  $n < 0$ . The cusp-regularity hypothesis of Proposition 5.10 is thereby removed.*

*Proof.* Write  $f = g + h$  as in Lemma 5.11:  $g \in M_k^!$ ,  $h$  is regular at the cusp with  $f$ 's interior poles, and both share a contour admissible for  $f$ . By linearity,  $L^*(f, s) = L^*(g, s) + L^*(h, s)$ . The cusp part is evaluated by Theorem 1.6 / (1.11) — valid for  $g \in M_k^!$  of any cusp order — giving  $-c_f(0)(1/s + i^k/(k - s))$  and the incomplete-gamma sum over the Fourier modes of  $g$ ; the cusp-regular  $h$  is evaluated by Proposition 5.10, contributing the interior polylog terms  $\sum_{\tau_p} \sum_m r_{f, \tau_p}^*(m) [I_{m-1}(s, \tau_p) + i^k I_{m-1}(k - s, \tau_p)]$ . Their sum is (5.12).  $\square$

**Remark.** The two parts are read at opposite fixed points:  $g \in M_k^!$  at  $\tau_0 = i$ , where the  $S$ -segment degenerates and (1.11) applies;  $h$  in the cusp limit  $\tau_0 \rightarrow i\infty$ , where the  $T$ -segment dies (Lemma 5.4(a)) and the deformed-Mellin form of [3] applies. No new integral is computed: (5.12) is a regrouping of (1.11) and the interior polylog terms. The split is available exactly when  $S_{2-k} = \{0\}$  (Lemma 5.11); at the exotic weights where it is not,  $L^*(f)$  is still finite (Lemma 4.2) and is taken directly from the finite contour.

## 6 Outlook

Two questions are left open. First, the geometric content of the bare Mellin kernel  $f(\tau) \tau^{s-1} d\tau$ : its integer- $s$  specialisation is the polynomial  $V_{k-2}$ -kernel  $(X - \tau Y)^{k-2}$ , but at general  $s$  we conjecture that  $\tau^{s-1} d\tau$  is a section of an infinite-dimensional principal-series local system on  $Y(1)$ , in the one-parameter family of the Lewis–Zagier period functions for Maass forms [13]; identifying it, and recasting the Hurwitz inversion of  $\delta_T$  as a coboundary statement there, is open. Second, a systematic per-mode evaluation of  $L^*$  on interior-pole forms parallel to (1.11), adapting the polylog projection  $\hat{P}_{\tau_p}$  of [3] to the finite-contour setting and matching the wall-crossing residues  $\mathcal{R}_{\tau_p}(f, s)$  to the iterated-Eichler-integral framework of [1]; the CM cases  $\tau_p \in \{i, \rho\}$ , where the Laurent data is arithmetic, are a clean target. Summed over a  $\Gamma$ -orbit, the residues (1.13) assemble into a weight- $k$  Poincaré-style sum in the pole location, taken up separately.

## A Numerical verification

The verification is layered through the proof of Theorem 1.4 as follows; the relevant named functions in `verify_periods.py` [4] are indicated in `typewriter`.

- **Cocycle integration** ((2.3), (2.7)): `cocycle_poly` computes  $C_S^f(\tau_0), C_T^f(\tau_0) \in V_{10} \otimes \mathbb{C}$  by  $V_{10}$ -valued contour integration along  $\gamma_{\tau_0}^{S,T}$ .
- **Coboundary inversion two ways** (Lemma 3.2): `solve_coboundary_LU` solves  $\delta_T P = (2\pi i)^{-(n+1)} C_T^f$  by forward substitution on the strictly-lower-triangular matrix (3.2); `solve_coboundary_bernoulli` uses the closed form (3.4) of Lemma 3.2. The two outputs agree to relative residual  $\sim 10^{-40}$ , confirming (3.4).
- **Brown-basis extraction at 9 interior monomials** (Lemma 3.7): `extract_periods` reads off  $\omega^\pm(\Delta)$  and  $\eta^\pm(\hat{\Delta})$  from  $r_f = C_S^f - (2\pi i)^{n+1}(P|_S - P)$  at each interior  $\ell \in \{2, \dots, 10\}$ . That all 9 agree to  $\lesssim 10^{-42}$ , matching the Brown values of Table 1, is the numerical check that the computed  $r_f$  lands on the cuspidal line (Corollary 3.8).
- **Basepoint independence** (Corollary 3.8): re-extraction of  $\eta^\pm(\hat{\Delta})$  at  $\tau'_0 = -0.2 + 0.9i$  agrees with the  $\tau_0 = 0.3 + 1.2i$  values to  $\lesssim 10^{-42}$ .
- **Clean-form identity** (1.9) (Theorem 1.4): `hurwitz_extract` evaluates the bare-Mellin + bare-Hurwitz integral form directly at  $\ell \in \{2, 3, 4, 5\}$  for  $f = \Delta$ , agreeing with the Brown reference values in Table 1 to  $\lesssim 10^{-42}$ .
- **$S$ - vs  $T$ -segment growth-rate cancellation** for  $f = \hat{\Delta}$ : `bare_integrals` confirms that  $|I_S|, |I_T|$  each grow  $\sim 10^7$ -fold across  $\text{Im}(\tau_0) \in [1.2, 2.5]$  while  $I_S + I_T$  stays  $\tau_0$ -invariant.

Analogous verification at higher weights  $k \in \{16, 18, 20, 22, 26\}$  where  $\dim S_k = 1$  requires only the Brown basis coefficients  $c_S^{\ell-1}$  at those weights and is structurally identical.

## B Algebraic derivation of the Hurwitz-zeta kernel

The body of Section 4.2 derives  $\tilde{k}_T(\tau, s)$  by differentiating the two-segment functional in  $\tau_0$  and demanding the result vanish. Here we record the parallel derivation that proceeds *from* the classical Mellin transform of a holomorphic cusp form, splitting it at the contour endpoints  $\{-1/\tau_0, \tau_0\}$  and

demanding the resulting tail be representable as a  $T$ -segment integral. Both derivations produce the same shift-difference equation and the same kernel; this appendix makes the algebraic provenance of the Hurwitz-zeta combination manifest.

**Step 1: Mellin split at the contour endpoints.** Let  $f \in S_k$  with  $k$  even. The classical completed  $L$ -function  $L^*(f, s) = (2\pi)^{-s} \Gamma(s) L(f, s)$  has integral representation

$$L^*(f, s) = -i \int_0^{i\infty} f(\tau) (\tau/i)^{s-1} d\tau = i^{-s} \int_0^{i\infty} f(\tau) \tau^{s-1} d\tau, \quad (\text{B.1})$$

the second equality using  $(\tau/i)^{s-1} = i^{1-s} \tau^{s-1}$  on the imaginary axis. Pick  $\tau_0 \in \mathbb{H}$  and split the imaginary-axis contour at the two endpoints  $\{-1/\tau_0, \tau_0\}$  in (1.5). Reorganising the three resulting pieces using  $S$ -modularity  $f(-1/\tau) = \tau^k f(\tau)$  on the lower segment  $[0, -1/\tau_0]$  (via the substitution  $\tau \mapsto -1/\tau$ ) and computing the  $q$ -series-against- $\tau^{s-1}$  integral on  $[\tau_0, i\infty]$  directly yields, for every  $\tau_0 \in \mathbb{H}$  and every  $s \in \mathbb{C}$ ,

$$L^*(f, s) = i^{-s} \int_{-1/\tau_0}^{\tau_0} f(\tau) \tau^{s-1} d\tau + \sum_{n \geq 1} c_f(n) \left[ \frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right]. \quad (\text{B.2})$$

The left side is  $\tau_0$ -independent (it is the classical  $L$ -value); both terms on the right side individually depend on  $\tau_0$ , but their sum does not. At  $\tau_0 = i$  the argument  $-2\pi i n \tau_0 = 2\pi n$  is real and (B.2) reduces to the standard BFK incomplete-gamma sum of [2].

The right side of (B.2) contains the  $S$ -segment integral  $i^{-s} \int_{-1/\tau_0}^{\tau_0}$  explicitly; the remaining incomplete-gamma sum plays the role that we want to be a  $T$ -segment integral.

**Step 2: ansatz for the  $T$ -segment kernel.** Postulate that there exists a kernel  $\tilde{k}_T(\tau, s)$ , analytic in  $\tau \in \mathbb{H}$  and meromorphic in  $s \in \mathbb{C}$ , such that

$$i^{-s} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = \sum_{n \geq 1} c_f(n) \left[ \frac{\Gamma(s, -2\pi i n \tau_0)}{(2\pi n)^s} + i^k \frac{\Gamma(k-s, -2\pi i n \tau_0)}{(2\pi n)^{k-s}} \right] \quad (\text{B.3})$$

for every  $f \in S_k$  and every  $\tau_0 \in \mathbb{H}$ . If such a kernel exists, substituting (B.3) into (B.2) gives the two-segment functional  $L^*(f, s; \gamma_{\tau_0}^S, \gamma_{\tau_0}^T) = i^{-s} [\int_{\gamma_{\tau_0}^S} + \int_{\gamma_{\tau_0}^T}]$  as a  $\tau_0$ -independent expression equal to  $L^*(f, s)$ .

**Step 3:  $\tau_0$ -derivative gives a shift-difference equation.** Differentiate both sides of (B.3) in  $\tau_0$ . On the left, the only  $\tau_0$ -dependence is in the integration limits, so  $d/d\tau_0$  acts as

$$\frac{d}{d\tau_0} \int_{\tau_0-1}^{\tau_0} f(\tau) \tilde{k}_T(\tau, s) d\tau = f(\tau_0) [\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s)], \quad (\text{B.4})$$

using  $f(\tau_0 - 1) = f(\tau_0)$  for the lower limit. On the right of (B.3), differentiating each incomplete gamma in its second argument (the explicit  $\tau_0$ -dependence) and assembling the Fourier series yields, after canceling a common factor of  $f(\tau_0)$ ,

$$\tilde{k}_T(\tau_0, s) - \tilde{k}_T(\tau_0 - 1, s) = -\tau_0^{s-1} + e^{i\pi(s-1)} \tau_0^{k-1-s} \quad \text{for all } \tau_0 \in \mathbb{H}. \quad (\text{B.5})$$

The first term on the right is the contribution of the  $\tau^{s-1}$ -Mellin endpoint at  $\tau_0$ ; the second term is the  $S$ -modularity image of the same endpoint at  $-1/\tau_0$ , carrying the branch factor  $(-1/\tau_0)^{s-1} = e^{i\pi(s-1)} \tau_0^{-(s-1)}$  together with the modularity Jacobian  $\tau_0^k$ . Equation (B.5) is a shift-difference equation of order one in  $\tau$ , with a sum of two power-of- $\tau$  terms on the right side.

**Step 4: Hurwitz raising solves the shift-difference equation.** The Hurwitz zeta function  $\zeta(\sigma, a)$  satisfies the raising identity

$$\zeta(\sigma, a+1) - \zeta(\sigma, a) = -a^{-\sigma} \quad (\sigma \in \mathbb{C}, \operatorname{Re}(a) > 0) \quad (\text{B.6})$$

(see [6, §25.11.3]). Applied at  $(\sigma, a) = (1-s, \tau)$  and  $(\sigma, a) = (s-(k-1), \tau)$  respectively, this gives

$$\zeta(1-s, \tau+1) - \zeta(1-s, \tau) = -\tau^{s-1}, \quad \zeta(s-(k-1), \tau+1) - \zeta(s-(k-1), \tau) = -\tau^{k-1-s}. \quad (\text{B.7})$$

Combining with the relative phase  $e^{i\pi(s-1)}$  from the second term of (B.5) gives the kernel

$$\tilde{k}_T(\tau, s) := \zeta(1-s, \tau+1) - e^{i\pi(s-1)} \zeta(s-(k-1), \tau+1), \quad (\text{B.8})$$

which by construction satisfies (B.5) and hence (B.3). The kernel is analytic in  $\tau$  on  $\mathbb{H}$  (both Hurwitz zetas have second argument  $\tau+1$  with  $\operatorname{Re}(\tau+1) > 0$ ) and meromorphic in  $s$  with simple poles only at  $s=0$  and  $s=k$ , where the per-mode integrals collect removable cancellations (see Proposition 4.4). Equation (B.8) is the kernel (1.7) of the main text.

**Where DR2022 stops.** The kernel (B.8) has two Hurwitz-zeta terms. The construction of [8, Thm. 1.2] gives the single-term

$$\tilde{k}_T^{\text{DR}}(\tau, m) = \zeta^*(1-m, \tau+1) \quad (\text{B.9})$$

on the segment  $[i, i+1]$ , at integer  $m$ , which is the first term of (B.8). The structural origin of this gap is the absence of an  $S$ -segment in their construction: without the  $S$ -modularity-transformed contribution at the  $-1/\tau_0$  endpoint in Step 3, the shift-difference equation (B.5) has only the single right-hand-side term  $-\tau_0^{s-1}$ , and its solution is the single-Hurwitz kernel. The second  $e^{i\pi(s-1)} \tau_0^{k-1-s}$  term appears precisely because the two-segment functional includes the  $S$ -segment and its endpoint at  $-1/\tau_0$  contributes under  $\tau_0$ -differentiation via  $S$ -modularity.

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- [2] K. Bringmann, K.-H. Fricke, and Z. Kent, *Special  $L$ -values and periods of weakly holomorphic modular forms*, Ramanujan J. **24** (2011), 119–139.
- [3] D. A. McGady,  *$L$ -functions for Meromorphic Modular Forms and Sum Rules in Conformal Field Theory*, arXiv:1806.09874 [hep-th] (2019); INSPIRE: <https://inspirehep.net/literature/1679799>.
- [4] Companion `mpmath` verification scripts (50-digit precision). `verify_periods.py` checks Theorem 1.4 at  $k=12$  for  $f \in \{\Delta, \hat{\Delta}\}$ : it reproduces Brown’s  $\omega^\pm(\Delta)$  and  $\eta^\pm(\hat{\Delta})$  [1], confirms basepoint independence and the closed form for  $\delta_T^{-1}$  (Lemma 3.2), and verifies the kernel identity of Lemma 3.5 symbolically. `dr_b_Lfunction_complex_s.py` checks Theorem 1.6 at non-integer  $s$  (half-integer, off-axis, and outside-strip points), reproducing  $L^*(\Delta, s)$  to  $\sim 10^{-53}$  and  $L^*(\hat{\Delta}, s)$  to  $\sim 10^{-47}$ , with  $\tau_0$ -independence. `verify_s6.py` and `verify_multi_s.py` check the per-mode  $\tau_0 = i$  form of Theorem 1.6 at  $s \in \{1, 3, 6, 9, 11\}$ , including the boundaries  $s=1, 11$  and the polar mode  $n=-1$ .

- [5] F. Brown, personal communication to the author, 28 June 2019. This correspondence describes the two-segment cocycle procedure (integrate along contours from  $\gamma^{-1}\tau_0$  to  $\tau_0$  for the two generators  $\gamma \in \{S, T\}$  of  $\Gamma$ , modify by a coboundary  $P|(1 - T)$  to kill the  $T$ -component, then read  $\omega^\pm$  and  $\eta^\pm$  off the  $S$ -component), and is the structural starting point of the construction in the present note. An accompanying unpublished note on quasi-periods of cusp forms of levels 6 and 8 was attached to the correspondence.
- [6] NIST Digital Library of Mathematical Functions, F. W. J. Olver *et al.*, eds., <https://dlmf.nist.gov/>, Release 1.2.x. Chapters used: §8 (incomplete gamma) and §25.11 (Hurwitz zeta).
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